

Testing the Simulation Hypothesis: A Critical Review of Arguments, Proposed Signatures, and Empirical Constraints

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Abstract

The hypothesis that our universe is a computer simulation is widely discussed but rarely stated in a form that can be tested. We review the philosophical arguments for it, its antecedents in digital physics, and every empirical signature proposed in the literature that we could identify. These signatures are a spacetime lattice, finite spatial resolution, render-on-demand quantum measurement, bounded computational resources, and information-physical effects. For each one we state the auxiliary assumptions it needs, the observations that constrain it, and the non-simulation physics that predicts the same signature. We argue that the unrestricted hypothesis cannot be refuted by any observation. Only conjunctions with explicit assumptions about the simulator’s architecture and resources, and about whether it adapts to being probed, can be tested. For every such sub-hypothesis we found, a detection would establish new physics, not simulation, and a null result bounds only a class of simulator designs. The review also contains four analyses of our own. First, applying current Lorentz-violation bounds to the lattice dispersion relations of Beane, Davoudi and Savage raises the lower limit on the inverse lattice spacing of their unimproved-Wilson scenario from $\sim 10^{11}$ GeV to $\gtrsim 10^{16}$ GeV. We also find that photon decay into e^+e^- is kinematically allowed far below the lattice scale, contrary to the original analysis. Second, we find no published search for the cubic-symmetric cosmic-ray anisotropy that scenario predicts, and we specify one that uses public data. Third, we show that the success criteria of the proposed render-on-demand experiments contradict standard quantum mechanics. Fourth, we reproduce the SARS-CoV-2 entropy values underlying the proposed “second law of infodynamics” and show that the reported decrease follows from the known mutational bias of the virus. Every factual claim in the paper is backed by a public ledger that records its source, location and supporting quotation.

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1 Introduction

Bostrom’s simulation argument [1] made the idea that we might live in a computer simulation a topic for academic philosophy, and it has drawn a large critical literature (Sec. 2). Physicists have since proposed ways to test the idea. They have looked for lattice artefacts in the highest-energy cosmic rays [2], proposed interferometric searches for Planckian “pixelation” of space [3, 4], and suggested quantum-optics experiments to detect a simulator that renders only what is observed [5]. Others have argued that computational complexity or undecidability rules simulation out [6, 7], or that an information-theoretic “second law of infodynamics” supports it [8]. These results have often been reported as proving or disproving the hypothesis, beyond what the papers themselves claim (Sec. 4.6).

The literature is scattered across philosophy, lattice field theory, astroparticle physics, quantum optics, complexity theory and thermodynamics, and we are not aware of a review that treats all of it together. This review has three aims:

1. to set out, for each proposal, what it assumes, what it derives, and what has actually been observed, with bounds, detections and proposals kept distinct;
2. to give a methodological framework that states precisely what any such test can and cannot establish (Sec. 4);
3. to identify where new work is possible, and to carry out the cheapest parts of it.

Position. We do not argue for or against the simulation hypothesis. Our thesis is methodological. The unrestricted hypothesis is not an empirical claim. Only its conjunctions with explicit assumptions about a simulator’s architecture, resources and non-adaptivity can be tested. Each such conjunction predicts signatures that non-simulation physics also predicts (Table 2). Throughout, we therefore speak of *constraining simulator designs*, never of proving or disproving the simulation hypothesis.

Contributions of this work. Besides the review itself, the paper contains analyses that are, to our knowledge, new. Each is labelled “this work” where it appears.

- Mapping the lattice dispersion relations of Beane et al. [2] onto published Lorentz-invariance-violation bounds. For their unimproved-Wilson scenario this gives $b^{-1} \gtrsim 10^{16}$ GeV from the Crab synchrotron spectrum [9], compared with the original $\sim 10^{11}$ GeV. We also show that $\gamma \rightarrow e^+e^-$ is kinematically allowed far below the lattice scale, and we determine how the bounds change under improved discretizations (Sec. 5.4).
- A documented search showing that no analysis of cosmic-ray arrival directions for a cubic-symmetric anisotropy has been published, and a specification of such an analysis using public Pierre Auger Observatory data (Sec. 6.7).
- An analysis showing that the success criteria of the proposed render-on-demand experiments [5] contradict standard quantum mechanics by up to 1/2 in probability (Sec. 9.2).
- A reproduction of the SARS-CoV-2 entropy values underlying the “second law of infodynamics” [10]. It shows that the reported decrease follows from the virus’s known mutational bias. We also give a calorimetric consistency test of the mass–energy–information conjecture (Sec. 10).

Verification protocol. Referees and readers should be able to check every statement. Each factual claim about the literature is recorded in a claim ledger. The ledger gives the source, a precise locator (section, equation, page or table) and a verbatim supporting quotation, and it records whether we read the full text, only the abstract, or only the bibliographic record. Claims we could not verify are excluded from the text. Bibliographic data were taken from DOI, INSPIRE-HEP or arXiv metadata. A script checks that every citation is backed by a verified ledger entry and that every DOI and arXiv identifier resolves to the cited title. The ledgers, derivations and reproduction scripts are released with the paper (Sec. 11.3).

Outline. Section 2 reviews the simulation argument and its critics, and Sec. 3 separates the simulation hypothesis from neighbouring “digital physics” theses. Section 4 gives the methodological framework. Sections 5–10 review the proposed signatures: lattice discreteness and Lorentz violation, ultra-high-energy cosmic rays, holographic noise, complexity and computability, render-on-demand quantum measurement, and information physics. Section 11 summarizes what is constrained and sets out open problems.

2 The simulation argument in philosophy and formal epistemology

Most scientific interest in whether the universe is a computation goes back to a probabilistic argument by Bostrom [1]. This section states that argument exactly, sets out the main lines of objection in the philosophical literature, and says what the argument does and does not imply for empirical work. That last point connects this section to the physical proposals reviewed later.

2.1 Bostrom’s argument

Premises. The argument has one metaphysical premise, one empirical premise and one epistemic premise. The metaphysical premise is a weak *substrate-independence* thesis. Bostrom does not require that it be necessarily true, only that “in fact, a computer running a suitable program

would be conscious”. He needs only that replicating a brain’s computational processes “in suitably fine-grained detail, such as on the level of individual synapses” would suffice for experience. He takes this “as a given” rather than arguing for it [1, § II].

The empirical premise concerns computing capacity. Bostrom estimates the processing needed to emulate one human brain at $\sim 10^{14}$ or $\sim 10^{16}$ – 10^{17} operations per second. He estimates the processing needed for an *ancestor-simulation* of the mental history of humankind at roughly 10^{33} – 10^{36} operations. The second figure is the product of 10^{11} humans, 50 years each, 3×10^7 s per year and 10^{14} – 10^{17} operations per second [1, § III and fn. 10]. He takes 10^{42} operations per second as a rough figure for a planetary-mass computer built from known nanotechnological designs. On that figure, one such computer could run an ancestor-simulation “by using less than one millionth of its processing power for one second” [1, § III]. The 10^{42} figure comes from an unpublished working manuscript that Bostrom cites. It is not derived from physical law.

The resource estimate depends on a further assumption: a simulation need only reproduce what the simulated observers can notice. Detail is filled in “on an as-needed basis”, and anomalies are handled by editing the brain states of the observers who noticed them or by re-running the simulation [1, § III]. Bostrom himself states that simulating “the entire universe down to the quantum level is obviously infeasible, unless radically new physics is discovered” [1, § III].

The fraction formula. Let f_P be the fraction of human-level technological civilizations that reach a posthuman stage. Let $\bar{N}$ be the average number of ancestor-simulations run by a posthuman civilization, and $\bar{H}$ the average number of individuals who lived in a civilization before it became posthuman. Bostrom writes the fraction of observers with human-type experiences who live in simulations as [1, § IV]

$$f_{\text{sim}} = \frac{f_P \bar{N} \bar{H}}{(f_P \bar{N} \bar{H}) + \bar{H}}. \quad (1)$$

Let f_I be the fraction of posthuman civilizations interested in running ancestor-simulations, and $\bar{N}_I$ the average number such civilizations run. Then $\bar{N} = f_I \bar{N}_I$ and

$$f_{\text{sim}} = \frac{f_P f_I \bar{N}_I}{(f_P f_I \bar{N}_I) + 1}. \quad (2)$$

Bostrom argues that $\bar{N}_I$ is extremely large, and concludes that at least one of the following holds [1, § IV]: (1) $f_P \approx 0$; (2) $f_I \approx 0$; (3) $f_{\text{sim}} \approx 1$.

The indifference step. A separate premise turns proposition (3) into a credence. This is the *bland indifference principle*, $\text{Cr}(\text{SIM} \mid f_{\text{sim}} = x) = x$. Bostrom says it “prescribes indifference only between hypotheses about which observer you are, when you have no information about which of these observers you are”. He supports it with an analogy to an unexpressed junk-DNA sequence and a betting argument [1, § V]. He also says it is weaker than the premise of the Doomsday argument, which he describes as “much stronger and more controversial” [1, § V].

What Bostrom claims. The conclusion is a disjunction, and Bostrom has repeatedly declined to endorse its third disjunct on its own. In 2003 he recommended apportioning credence “roughly evenly” among (1)–(3) [1, § VII]. In 2005 he wrote that “the disjunction is all that the simulation argument purports to show” [11, § 1]. In 2009 he wrote: “I believe that we are probably not simulated” [12]. His current FAQ assigns a “substantial probability” to the hypothesis but gives no number [13, Q2]. Bostrom also held that the truth of (3) would leave most beliefs nearly intact,

since “our best guide” to how the simulators set up our world “is the standard empirical study of the universe we see” [1, § VI].

The 2011 patch. Bostrom and Kulczycki later identified “a mathematical non sequitur” in Eq. (1) [14]. The formula uses one average $\bar{H}$ for all civilizations. Suppose instead that civilizations which go on to run simulations have much shorter pre-posthuman phases than those that do not. Then all three disjuncts can fail at once. The paper offers two independent patches. The first assumes that cumulative pre-posthuman populations do not differ by an astronomically large factor between simulating and non-simulating civilizations; its appendix uses a factor of one million. The second restricts the reference class using indexical evidence about our position in history, such as a “computer age birth rank”. Throughout, the paper assumes a finite universe [14].

2.2 Objections to the inference

The indifference step and the nature of evidence. Weatherson [15] distinguishes four readings of the general principle behind the indifference step. On two readings, he argues, it is false. On the third it does not yield Bostrom’s principle. On the fourth it yields the principle only given an auxiliary premise: that all of one’s evidence is probabilistically independent of being a Sim, which “we have no reason to believe”. Against that premise he raises epistemological externalism: “seeing a dagger and sim-seeing a sim-dagger” may give different evidence even if the experiences match. He also raises a Goodman-style challenge built on the gerrymandered predicate “suman” [15].

Bostrom replied that the principle constrains *conditional* credence, so that further evidence enters by conditionalization. On his reply, the argument requires only that our actual evidence is too weak to move credence far once $f_{\text{sim}} \approx 1$ is given [11, § 2]. He also argued that the simulation argument differs from brain-in-a-vat scepticism. It relies on empirical premises and aims to raise credence in a disjunction about the world, not to produce general agnosticism [11, § 4].

Birch [16] makes explicit the constraint that the argument places on credence: $\text{Cr}(\text{SIM}) \geq \text{Cr}(\text{SIM} \mid H_3) (1 - \text{Cr}(H_1 \vee H_2))$. He then argues that the argument needs a “selective scepticism”. The trilemma requires good evidence about the physical limits of computation. The indifference step requires that our evidence not support mundane claims about our own constitution. Given a “parity of evidence” premise, these requirements are “jointly incompatible”. Retreating to a weaker four-way disjunction does not recover the constraint on credence. Birch also argues that a simulated observer could infer with confidence only that at least one ancestor-simulation is possible. He concludes that “there is thus no good reason to believe” the argument’s conclusion [16].

Crawford (abstract-level) gives a structurally parallel argument that one is a “freak observer” formed in a fluctuation. He also argues that “the evidence that simulants or freak observers exist is not a reason to think that I am one of them” [17].

Thomas [18] accepts that Bostrom’s reasoning establishes only a conditional claim. He notes that confidence in a high ratio of simulants is hard to square with being a simulant. He therefore replaces that premise with a weaker one: the expected simulant-to-non-simulant ratio in one’s reference class is high, *conditional on* one’s being non-simulated. Given an indifference principle and an admissibility condition on the reference class, he proves that the odds of being simulated are at least this expected ratio. He also argues that finding many simulations being run would not necessarily raise those odds [18]. This runs against Bostrom’s remark that our going on to create ancestor-simulations “would be strong evidence against (1) and (2)” [1, § VI].

Substrate-independence. Some objections target the metaphysical premise. Beisbart (abstract-level) argues that the interactions among hardware components during a brain simulation “do not

resemble the ways in which neurons interact in the brain” [19]. Summers and Arvan (abstract-level) argue that if panpsychism or panqualityism is true, living in a simulation might be possible only as a brain in a vat, which would make simulation unlikely [20]. Chalmers reports that Godfrey-Smith [21] favours a biological basis for experience, and replies that functionalist views remain serious possibilities [22].

The motivational disjunct. Bostrom noted that an ethical ban on ancestor-simulations would support (2) only if it were combined with a civilization-wide means of enforcing it [1, § VI]. Greene [23] argues that running ancestor-simulations, and probing experimentally for simulation, carry a risk that the simulators terminate our world. Harris [24] builds on this. He argues that any beings unsure whether they are simulated have self-preservation reasons not to run conscious simulations, since a nested simulation could exhaust the resources of a lower level. This supports disjunct (2), and Harris concludes that a high probability for (3) “is premature” [24].

2.3 Nesting and resource decay

Bostrom allowed that simulated civilizations might run simulations of their own. He also noted that the cost to the basement level counts against many levels, so that “we should expect our simulation to be terminated when we are about to become posthuman” [1, § VI]. Brueckner objected that a simulated being cannot “really” build another computer [25]. Our knowledge of this objection comes from the opening paragraph of Brueckner’s paper and from Bostrom’s quotation of it. Bostrom replied that virtual machines really perform the computations they emulate [12].

Carroll [26] states a “resolution conundrum”. Each level has less computing power than the one above it, so the hierarchy ends in simulations unable to simulate. With a typicality premise, most observers then live at that lowest level. This conflicts with the premise that such simulations are feasible. Carroll calls the typicality premise “more or less completely without justification”. Kipping’s hierarchical model agrees that most realities lie in the lowest level when $p\lambda \gg 1$ (notation below). He suggests that inhabitants of the lowest level might still build detailed, non-sentient simulations and so reach the same trilemma [27, § 2.3]. Bostrom’s FAQ proposes that simulators could cap simulated civilizations’ computing, which would make most civilizations “leaf nodes of the simulation tree” [13, Q6].

Bibeau-Delisle and Brassard write a Drake-style equation for the simulated fraction. In a recursive scenario they bound it above by a simulation-efficiency factor that they argue “is most likely under 50%” [28].

These authors draw different conclusions from one observation: our civilization has not produced simulated minds. This comparison is ours, not any single author’s. Richmond takes the observation to make our level unusual on Bostrom’s picture, which would favour hierarchies with fewer levels [29]. Lewis shows that in some models conditioning on it *raises* the credence that one is simulated [30]. Kipping uses it to obtain a Bayes factor close to unity [27].

2.4 Self-locating belief and anthropic reasoning

The indifference step is one instance of a wider problem: how to assign credence when one is uncertain which observer one is. Bostrom’s *Self-Sampling Assumption* (SSA) says that one should reason as if one were a random sample from all observers in one’s reference class [31, p. 57]. He later restates it in terms of observer-moments. The rival *Self-Indication Assumption* (SIA) favours hypotheses on which many observers exist. Bostrom rejects SIA, mainly through the “Presumptuous Philosopher” case [31, pp. 66, 124–125]. He also shows that whether an entity belongs to the

reference class can change the rational credence when population size depends on the hypothesis [31, pp. 69–71].

Elga defends the “thirder” answer to Sleeping Beauty [32]. He also defends an indifference principle: “similar centered worlds deserve equal credence” [33]. On Elga’s view, hypotheses about one’s location among duplicates carry no default bias against them, whereas ordinary sceptical hypotheses do [33]. Birch uses this distinction to specify the rare circumstances in which Bostrom’s selective scepticism could be defensible, and argues that we are not in them [16].

Lewis [30] argues that the simulation argument succeeds where the Doomsday argument fails. His result depends on a location-uniform prior, which he identifies with the thirder position. Richmond (abstract-level) argues the opposite: anti-simulation conclusions follow with “more ... robust reference classes” and without indifference principles [34].

Two limitations run through this literature. First, it mostly assumes finitely many observers [14]. Bostrom’s FAQ notes that in an infinite universe the simulated fraction “is not defined”, and proposes a limit-density measure [13, Q7]. Second, the verdict depends on the choice of reference class and self-sampling rule, and that choice is itself disputed.

2.5 Quantitative treatments

Kipping [27] combines disjuncts (1) and (2) into a “physical hypothesis” H_P . He compares it with a simulation hypothesis H_S in which a base civilization runs λ simulations, each of which runs its own simulations with probability p , down to a maximum depth G . Assigning equal prior probability to H_P and H_S , and conditioning only on one’s existence, he obtains a probability of being in base reality of $\frac{1}{2} + 1/[2(N_{\text{sim}} + 1)]$, where N_{sim} is the total number of simulated realities. The probability of being simulated is therefore below 50% and tends to 50% as $N_{\text{sim}} \rightarrow \infty$. Conditioning instead on the fact that we have not produced simulations gives a Bayes factor of $(\lambda - 1)/\lambda$. Kipping also finds that if humanity began producing such simulations, the probability of base reality would fall to λ/N_{sim} [27].

The 50% ceiling comes from the equal-prior assumption, not from data. Kipping himself notes that equal priors could be “too generous” to H_S because it is the more complex model [27, §3]. The v1 text also contains two prose sentences that contradict its own equations; we rely on the abstract and equations. The Kipping and Bibeau–Brassard models [27, 28] make Bostrom’s framework explicit. The numbers they produce follow from assumed priors, branching structure and efficiency factors, none of which is measured.

2.6 Simulation and scepticism

Chalmers argues that the hypothesis that one is in a matrix “is not a skeptical hypothesis, but a metaphysical hypothesis” [35]. He treats it as equivalent to three claims: our world was created, its microphysics is computational, and our minds are constituted outside it. If this is right, most ordinary beliefs remain true. Local or recent simulations are at worst “partial skeptical hypotheses” [35].

In *Reality+* [36], as summarized in his précis [37], Chalmers restates the argument as “if there are no sim blockers, we are probably sims”. His sim blockers include Bostrom’s two disjuncts and also the impossibility of intelligent or conscious sims, simulators’ avoidance of conscious sims, and excessive computing cost. He argues that we cannot know that any sim blocker obtains, so we cannot know that we are not simulated [37]. His anti-sceptical move is structuralist: a simulation with the same causal structure as an unsimulated universe makes the same physical theories true [37]. Replying to critics, he gives an illustration: 10% credence in the denial of each of two sim blockers

leaves “a 1% credence that most humanlike beings are sims”. He rejects global scepticism but leaves local scepticism open [22].

Schwitzgebel (abstract-level) argues that, conditional on being simulated, we should give significant credence to a small or brief simulation [38]. Summers and Arvan argue that, under panpsychism or panqualitism, viable simulation hypotheses become “substantially sceptical” [20].

Ethics and theology. In peer-reviewed philosophy of religion, Dainton proposes a “simulation solution” to the problem of natural evil [39]. Crummett assesses it against Fall and diabolical theodicies [40]. Johnson argues that some theists are committed to our world most likely being a simulation [41], and Steinhart draws out further theological analogues [42]. All four are cited at abstract level.

2.7 What the argument does and does not license empirically

Five points link this section to the physical proposals reviewed in later sections.

1. *The argument does not predict any observable signature.* Its conclusion is a disjunction whose third disjunct concerns the fraction of observers who are simulated, not the physics they observe [1, 11]. Bostrom’s resource estimate assumes that simulators render only what observers notice [1, §III]. His FAQ doubts that lattice-artefact searches of the kind proposed by Beane *et al.* would detect a simulation. It gives two reasons: simulators need not use “the crude simulation technique that such a test would detect”, and they “could fake the results” [13, Q12]. Any physical test must therefore add assumptions about how a simulator would be built. Those assumptions are not supplied by the argument.
2. *Observations bear on it only indirectly.* Bostrom holds that the hypothesis is testable only in the weak sense that evidence bearing on disjuncts (1) or (2) shifts credence in (3) [13, Q11]. Qualitative expectations do exist in the literature. For example, Barrow suggests occasional “glitches” and slow drifts in the constants [43]. These are motivated guesses, not consequences of the argument.
3. *Null results and positive results are asymmetric.* Greene argues that no experiment could show that we are *not* simulated, since a simulator could program any observation; only a positive anomaly would be diagnostic [23]. Bostrom’s FAQ makes a related point from the other direction: philosophical arguments, but not experiments, are immune to the concern that a successful test might prompt termination [13, Q12].
4. *Physics constrains the premises, not the conclusion.* Empirical bounds on the cost of simulating physics bear on the empirical premise about computing capacity, that is, on the “sims will require too much computer power” blocker [37]. But Birch argues that a simulated observer’s physics need not reflect the true limits of computation outside the simulation [16]. Bostrom’s FAQ replies with a two-case argument [13, Q4]. Neither side’s reasoning is settled by data.
5. *Credences in the literature rest on priors.* Stated probabilities range from Bostrom’s “roughly even” split [1], to “probably not simulated” [12], to Kipping’s “less than 50%” [27]. Each follows from contested indifference principles, reference classes or model priors. None is a measurement.

3 Antecedents and neighbouring programmes: digital physics and computational views of nature

The simulation hypothesis is often presented as the latest form of an older idea, that “the universe is a computer”. That older literature contains several different theses. If they are run together, it becomes unclear what a given observation could test. The key distinction for this review is between the claim that the universe *is* a computation (or is computable, or is discrete) and the claim that it is *run on* a computer that belongs to some other reality. This section traces where the computational views came from and fixes the terms used in later sections.

3.1 Six theses that are often conflated

We separate six claims. Table 1 summarizes them.

(C) *Computational modellability*. Floridi separates two questions. The first is whether the physical universe can be adequately modelled digitally and computationally. The second is whether it is digital and computational in itself. He treats the first as an open “empirico-mathematical” question and argues that the second is ill-posed [44]. Thesis (C) carries no ontological commitment. Floridi puts this as a contrast between the “predicative” and “attributive” senses of “digital physics”.

(CU) *Computable universe*. Szudzik defines a computable physical model as a recursive set of states together with a total recursive function for each observable. On this basis he states a *computable universe hypothesis*: the universe has a recursive set of states, and every observable is a total recursive function of the state [45]. His examples include a model with non-discrete, continuous motion. So computability in this sense does not require discreteness. Tegmark’s *Computable Universe Hypothesis* is a different formulation. It requires that the relations defining the mathematical structure identified with our universe be computable by halting computations [46]. Deutsch’s physical reading of the Church–Turing thesis is a related but separate claim. It says that “every finitely realizable physical system can be perfectly simulated by a universal model computing machine operating by finite means” [47]. Deutsch argued that classical continuum physics does not satisfy this strong form with respect to the Turing machine. Quantum theory, he argued, is compatible with it through a universal quantum computer. He also held that the principle has the epistemological status of other physical principles, such as the third law of thermodynamics [47]. Deutsch himself calls it “the Church–Turing principle”.

(D) *Digital universe*. This is the thesis that the physical universe (time, space and every entity and process in spacetime) is ultimately discrete. Floridi takes it as the core of “digital ontology” [44]. The SEP entry on physical computation adds a requirement for a cellular-automaton universe. All fundamental magnitudes must take finitely many values, and space and time must be fundamentally discrete or emerge from discrete processing [48].

(P) *Ontic pancomputationalism*. Here the universe is itself a computing system. The SEP entry notes that this view originated in physics, not philosophy. It combines an *empirical* claim with a *metaphysical* one. The empirical claim is that fundamental magnitudes and their state transitions are *exactly* described by some computational formalism. The metaphysical claim is that computation is what the universe is made of [48]. There is a classical version, based on cellular automata, and a quantum version associated with Lloyd [49]. The entry describes the quantum version as possibly a reformulation of quantum mechanics without change in empirical content [48].

(M) *Mathematical universe*. Tegmark’s Mathematical Universe Hypothesis states that “our physical world is an abstract mathematical structure” [46]. It generalizes his earlier “ultimate ensemble” postulate that all structures that exist mathematically also exist physically [50].

Table 1: Computational theses about the physical world. The last column states whether the thesis itself posits computing hardware outside our universe.

Thesis	Core claim	Representative sources	External hardware?
(C) Modellability	The world can be adequately modelled computationally	[44]	No
(CU) Computable universe	States and observables (or defining relations) are computable	[45–47]	No
(D) Digital universe	Spacetime and all magnitudes are fundamentally discrete	[44, 51]	No
(P) Ontic pancomputationalism	The universe is a (classical or quantum) computing system	[48, 52, 53]	Not necessarily
(M) Mathematical universe	The world is a mathematical structure	[46]	No
(S) Simulated universe	The world is a computation run on hardware outside it	[1, 54]	Yes

(S) *Simulated universe*. This is the claim that our universe is a computation *implemented on hardware that exists outside it*. Usually the hardware is also assumed to have been built by designers. This is the simulation hypothesis proper, whose philosophical formulation is reviewed in the section on the simulation argument (Sec. 2).

Several relations among these theses are stated explicitly in the literature. First, (D) and (P) are independent. Floridi cites Wheeler as someone who held the former but not, at least not explicitly, the latter [44]. Second, (CU) does not entail (D), as Szudzik’s continuous example shows [45]. Third, (P) does not entail (S). The SEP entry notes that the metaphysical claim of (P) needs an account of what the computation *is*. It lists three options: a simulation running on a computer in another universe, computations as abstract mathematical entities (a computational Pythagoreanism), and computational structural realism [48]. Only the first is (S). Fourth, Tegmark contrasts his hypotheses with (S) explicitly. The Computable Universe Hypothesis “requires the description (the relations) rather than the time evolution to be computable”, so that “the universe is a mathematical structure according to the MUH, as opposed to a computation according to the simulation hypothesis” [46].

3.2 A genealogy

Zuse and Fredkin. Zuse’s “Rechnender Raum” first appeared as a journal article in 1967 [55]. It was followed by a book in 1969 [56] and an MIT technical translation, *Calculating Space*, in 1970 [57]. We could not consult Zuse’s text, so we rely on secondary characterizations. The SEP entry credits Zuse and Edward Fredkin with the “earliest and best known” version of ontic pancomputationalism, in which the universe is a giant cellular automaton. It adds that Fredkin’s largely unpublished ideas influenced Feynman, Toffoli and Wolfram [48]. Floridi calls the shared position the “Zuse Thesis”: the universe is deterministically computed on a giant but discrete computer. He notes that Zuse, Fredkin and Wolfram took that computer to be a cellular automaton, Schmidhuber a universal Turing machine, and Lloyd a quantum computer [44]. Fredkin developed “digital mechanics” [58] and “digital philosophy” [51]. As quoted by Floridi, digital philosophy takes space and time to be discrete and the bit as the unit of state. It also demands the “eventual ability to derive” the equations of current physics from such models [44]. According to the SEP entry, Fredkin [51] also considered the possibility that our universe is a simulation on a computer in another universe whose physics is inaccessible to us [48]. That possibility is thesis (S), not (D).

The MIT physics-of-computation papers (1981–1984). Feynman’s keynote “Simulating physics with computers” says that his interest in the topic was inspired by Fredkin [59]. The part that concerns us here is not complexity, which is treated in the section on computational complexity (Sec. 8), but Feynman’s notion of *exact* simulation. Exact simulation would require “everything that happens in a finite volume of space and time” to be “exactly analyzable with a finite number of logical operations”. Present physics does not meet this condition, so “if this proposition is right, physical law is wrong” [59]. Feynman also noted that if space were replaced by a simple lattice, the speed of light would generically depend slightly on direction. The resulting anisotropies are experimentally constrained [59]. This observation anticipates the lattice tests reviewed in the section on lattice signatures and Lorentz violation (Sec. 5). He called Fredkin’s programme of finding a computer simulation of physics “an excellent program”, provided the simulation is quantum mechanical [59]. Fredkin and Toffoli’s conservative logic went the other way, from computation to physics. It is a model of computation built to respect reversibility and conservation laws, and it shows that a perfect gas in a suitably shaped container can reproduce a general-purpose digital computer [60]. Toffoli later proposed cellular automata “as an alternative to (rather than an approximation of)” differential equations in modelling physics [61]. Zuse [62] and Wheeler [63] also published papers in the 1982 issue of the *International Journal of Theoretical Physics* that carried Feynman’s keynote.

Wheeler: “it from bit”. Wheeler’s slogan is often grouped with digital physics, but its content is different. It holds that every physical quantity “derives its ultimate significance from bits, binary yes-or-no indications”. These bits are elementary acts of “observer-participancy” [64]. Wheeler denied a microscopic continuum of space and time (“continuum-based physics, no; information-based physics, yes”). He also explicitly rejected “the concept of universe as machine”, partly because it requires postulating “a supermachine” [64]. “It from bit” is therefore an informational and participatory thesis close to (D). It is not a thesis of type (P) or (S).

Lloyd: limits and the computing universe. Lloyd derived that a system of mean energy E can perform at most $2E/\pi\hbar$ elementary operations per second. For a 1 kg “ultimate laptop” the bound is about 5.4×10^{50} operations per second [65]. Applied to the observable universe, the same limits give order-of-magnitude upper bounds of about 10^{120} operations on about 10^{90} bits (about 10^{120} bits if gravitational degrees of freedom are included). The operation count scales as $(t/t_P)^2$ [52]. Lloyd offered three readings of these numbers. They bound the computation that matter can have performed. They bound from below the resources a quantum computer needs to simulate the universe directly. And, in a reading he calls “more controversial”, they give the memory and operation count of the universe regarded as a computation [52]. The second reading bears on (S). The third is a quantum version of (P). Lloyd’s own answer to “Is the universe a computer?” depends “on the meaning of ‘computer’ and on the meaning of ‘is’”: the universe is not a conventional digital computer, but “it can still compute” [52]. A later preprint proposes deriving spacetime geometry from an underlying quantum computation [66].

Wolfram: *A New Kind of Science* and the 2020 models. Wolfram asked whether “some simple program which, if run for long enough, would reproduce our universe in every detail” might underlie physics. Such a program would be a model “not in any sense an approximation or idealization” [53]. In the same book he wrote that no literal mechanistic model of the Zuse–Fredkin kind “can ever in the end realistically be expected to work” [53, p. 1026]. Expert reviews of the physics chapter were critical. Aaronson argued that its causal-network picture of spacetime had been discussed earlier in quantum gravity, with the difference that Wolfram’s model is explicitly

classical. He also showed that Wolfram’s deterministic “long-range thread” account of entanglement cannot satisfy causal invariance, the relativity postulate and Bell-inequality violation all at once [67]. Kadanoff attributed the automaton-universe view to Fredkin. He judged the chapter “a partially formed idea—exciting, but not yet science” and found “no comparison with experiment” [68]. Weinberg’s essay review saw no motivation for the discrete-universe speculation beyond familiarity with computers. He also noted that the computational equivalence of automata to continuous systems holds only if nothing in nature is truly continuous [69]. In 2020 Wolfram introduced models in which the universe is an evolving hypergraph with no intrinsic space, time or matter. He proposed that “causal invariance” of the rewriting rules yields Lorentz invariance, general covariance and local gauge invariance, and described parts of this correspondence as “speculative” [70]. In that paper experimentally accessible predictions are a prospect (“it seems increasingly likely that experimentally accessible predictions will be possible”), not a result. The models’ elementary length need not coincide with the Planck length [70]. Gorard reports proofs that causal invariance is equivalent to a discrete general covariance, and derives discrete analogues of Lorentz covariance and, under a dimension-preservation assumption, of the Einstein equations [71]. He also relates the models to causal sets [72]. Butterfield and Dowker argue that any Planck-scale discrete theory that recovers general relativity must contain causal sets. They list the Wolfram model among the approaches whose status under this argument “it would be a good project to assess” [73]. The 2020 models allow multiway (branching) evolution, which the NKS framework did not. We did not find a published analysis of whether Aaronson’s argument carries over to them.

’t Hooft: the cellular automaton interpretation. ’t Hooft treats quantum mechanics “as a tool, not as a theory”. He suggests that the Standard Model with gravity may be a quantum-mechanical analysis of a system that “could be classical at its core”. He also argues that the usual objections to superdeterminism can be overcome “at least in principle” [74]. This is a (D)-type proposal about the foundations of quantum mechanics. It does not posit a simulator. It makes a “firm prediction”: quantum computers will not outperform a classical computer with one memory site per Planck volume (or Planck area) that performs one operation per Planck time [74]. This links the proposal to the complexity questions treated in the section on computational complexity (Sec. 8).

Algorithmic ensembles and the mathematical universe. Schmidhuber cast his 1997 analysis explicitly as a simulation: a “Great Programmer” runs all computable universes on “His Big Computer”. Here “computable” means discrete time and finitely describable states, and computing all universes needs far less information than computing one arbitrary universe [75]. He noted that inhabitants “run on local time” and cannot tell how many machine steps each of their time steps takes [75]. His later “speed prior” assigns low probability to histories that are slow to compute. From it he derived a concrete prediction: apparently random events such as individual beta decays come from a fast pseudorandom generator [76]. Tegmark separates his computable-universe hypothesis from (S), as quoted above. He also calls “a common misconception in the universe simulation literature” the identification of physical time with the step-by-step flow of a computation. On his account, a simulating computer need only *specify* a history, not compute it [46]. He concedes that “virtually all historically successful theories of physics violate the CUH” because they use the continuum [46]. Deutsch’s and Feynman’s observations point to the same tension [47, 59].

Early simulated-world proposals. In a 1992 essay Moravec argued that a future cyberspace would replay human history many times. Hence “the very moment we are now experiencing may

actually be (almost certainly is)” such a replayed event [54]. This informally anticipates the counting reasoning of the simulation argument [1]. Bostrom’s paper cites Moravec’s *Mind Children* [77] and Lloyd’s limits paper [65]. Tegmark lists Tipler [78], Bostrom and Schmidhuber as authors who discussed the probability that we are simulated [46]. He describes Lloyd’s view as an intermediate case: an analog quantum simulation “not designed by anybody” [46].

3.3 Philosophical assessments

The SEP entry distinguishes “unlimited” pancomputationalism from “limited” pancomputationalism. The unlimited form, associated with Putnam and Searle, holds that every sufficiently complex system implements very many computations. The limited form holds that every system implements some computation. Both differ from the ontic pancomputationalism of the physicists discussed here [48]. The entry finds “little positive evidence” for ontic pancomputationalism. It suggests that supporters are motivated by “the desire for exact computational models of the world” rather than by evidence. It notes the metaphysical objection that computational entities lack the causal and qualitative properties we observe. It also grants that the classical version is “in principle” testable [48]. Floridi argues that digital ontology should be abandoned in favour of informational structural realism. His reason is that “digital” and “analogue” are modes of presentation relative to a level of abstraction, not properties of reality in itself [44]. His argument has drawn at least one published reply [79].

3.4 Consequences for this review

These distinctions have three practical consequences for the rest of the review. The first two are our own inferences from the sources above; the third restates positions in them. First, empirical searches for spacetime discreteness (section on lattice signatures) test particular (D)-type models, such as a lattice with preferred directions. Their bearing on (S) depends on extra assumptions about how a simulator would represent our world. Nothing in (S) as such fixes a lattice representation. It has also been argued that the simulator’s steps need not correspond to our time, and that a simulator might only specify our history rather than compute it step by step [46, 75]. Second, the computability and complexity results (section on complexity) bear on (CU) and on the resource side of (S). Examples are Deutsch’s principle, Lloyd’s bounds (read as lower bounds on the cost of a direct quantum simulation) and ’t Hooft’s prediction about quantum computers [47, 52, 74]. Third, arguments that the universe “is” computational do not by themselves support the claim that it is simulated, because a computational ontology can be read in ways that posit no external hardware [46, 48].

4 The scientific status of the simulation hypothesis

This section sets out the methodological framework for the rest of the review. We ask what it would mean to test the claim that the observable universe is a computer simulation, which versions of the claim can be tested, what a positive or null result would show, and how published results have been described to the public. Our conclusion is that only restricted sub-hypotheses about how a simulator is built can be tested. For every sub-hypothesis we examine, the proposed signature is also predicted by non-simulation physics.

4.1 Standards of testability

Falsifiability. On Popper’s demarcation criterion, a theory is unscientific if it is compatible with all possible observations. This holds whether the theory is compatible by construction or because it has been modified to accommodate them [80]. Corroboration counts only when it comes from a “risky” prediction that might have failed [80]. Popper also held that a theory rescued from failed predictions by ad hoc hypotheses degenerates into dogma [80]. Lakatos’s standard objection is that high-level theories are always protected by a “protective belt” of auxiliary hypotheses. On his account they are abandoned when a research programme stalls, not after a single critical test [80].

Underdetermination. Stanford distinguishes two forms of underdetermination [81]. In the holist form, a failed prediction can be blamed on an auxiliary assumption rather than on the hypothesis under test. In the contrastive form, rival theories are equally well supported by the same evidence. In the limiting case of empirically equivalent theories, no possible evidence can favour one over the other [81]. Descartes’ evil demon is the classic sceptical instance: all sensory experience would be the same whether it came from the demon or from an external world [81].

Bayesian confirmation. Evidence e confirms a hypothesis h relative to background k if and only if $P(h \mid e \wedge k) > P(h \mid k)$. It is neutral if the two are equal [82]. One standard measure of how far e favours h over its negation is the likelihood ratio $P(e \mid h \wedge k) / P(e \mid \neg h \wedge k)$ [82]. We use this measure below. A signature discriminates between a simulation sub-hypothesis and a non-simulation rival only if their likelihoods for it differ.

Graded falsifiability and non-empirical assessment. Carroll argues that falsifiability tracks two virtues, definiteness and empiricism. On definiteness, a theory should not be compatible with “any conceivable set of facts” [83]. He ranks theories from category 1, where no conceivable test could return an incompatible result, to category 5, where there are doable and definitive tests. Category 4 covers theories that can be tested only over part of their parameter space [83]. On his view the important divide lies between category 1 and the rest. Category-1 theories “don’t really even have a chance of being true”. Ideas in category 4 are tested where possible, and interest in them fades as they are squeezed into smaller regions of parameter space [83].

A separate debate concerns whether theories may be assessed without empirical data. Dawid argues that non-empirical assessment plays an important part in high-energy physics [84]. He identifies three arguments: the no-alternatives argument, the meta-inductive argument from past success, and the argument from unexpected explanatory interconnections. He casts all three as Bayesian confirmation [85]. Ellis and Silk argued that exempting speculative theories from experimental verification undermines science [86]. Rovelli argues that non-empirical evidence cannot move a theory from “maybe” to “reliable”. He adds that no-alternatives arguments hold only under assumptions that may be false [87].

The simulation-hypothesis literature is largely non-empirical in Dawid’s sense. The arguments for it are probabilistic (Sec. 2), and several arguments against it are a priori or concern in-principle cost (Sec. 8). The no-alternatives argument is plainly unavailable to proponents, because each proposed signature has a non-simulation explanation (Sec. 4.3).

4.2 The unrestricted hypothesis

Let H_{SIM} denote the unrestricted claim that the observable universe is the output of a computation run by a system outside it. H_{SIM} places no constraint on the simulator’s hardware, algorithm,

resources or intentions. Chalmers argues that nothing could prove we are not in a simulation, “because any such evidence could be simulated” [88]. This passage is a publisher-authorized excerpt of Ref. [36]. Wolpert gives a precise example of implementation-level equivalence. Beings simulated through fully homomorphic encryption could not tell their situation apart from a direct simulation of their laws of physics [89]. A perfect simulation of a physical theory T and T itself are therefore empirically equivalent in the sense of Ref. [81]. Their likelihood ratio is unity for every possible observation. H_{SIM} is thus an underdetermination of the sceptical, Cartesian kind rather than an ordinary scientific one. In Carroll’s scheme it belongs in category 1. This placement is our application; Carroll discusses the multiverse, not simulation.

Two qualifications matter. First, the hypothesis is not symmetric under evidence. Chalmers holds that we could obtain “very strong evidence” that we are simulated, for instance if the simulators showed us the source code, even though this would fall short of proof [88]. Hsu and Zee discuss a physical channel for such a disclosure by a creator: a message encoded in the cosmic microwave background through the choice of fundamental Lagrangian [90]. H_{SIM} can therefore in principle be confirmed by a cooperative simulator, but it cannot be refuted. Second, Bostrom himself does not claim empirical content for the unrestricted hypothesis. He writes that the best guide to how simulators set up our world is “the standard empirical study of the universe we see”. He also writes that the chief empirical importance of the simulation disjunct lies in its role in the trilemma [1]. The probability assigned to H_{SIM} therefore comes from the prior, supplied by the simulation argument (Sec. 2), and not from data.

4.3 Restricted sub-hypotheses and their degenerate rivals

A test becomes possible only when H_{SIM} is conjoined with auxiliary assumptions A about the simulator. Beane, Davoudi and Savage make these assumptions explicit. They assume a classical computer, a hyper-cubic lattice and an unimproved Wilson action. From the cosmic-ray spectrum they derive $b^{-1} \gtrsim 10^{11}$ GeV, and they propose a search for lattice-aligned anisotropy in the highest-energy cosmic rays [2]. Table 2 classifies the restricted hypotheses discussed in this review. For each class it lists what would count as evidence and which non-simulation physics predicts the same signature.

Two general points follow from the table. The first is degeneracy. Suppose a sub-hypothesis $H_{\text{SIM}} \wedge A$ and a physical model P both predict a signature s . Then the likelihood ratio between them for s is close to unity. Observing s would confirm their disjunction relative to standard physics, but it would not favour simulation over P [82]. A detected lattice anisotropy would show that spacetime is discrete in a particular way. A detected deviation from linear quantum dynamics would support a collapse-type modification. In neither case would the result show that the discreteness or the collapse was computed by someone. For a simulation-specific reading, a signature would need features that the rival models do not motivate. An example is a discreteness scale far below the Planck scale, which Beane et al. note is possible in the simulation scenario [2]. Even then, contrastive underdetermination allows a non-simulation model to be built that reproduces the feature [81].

The second point concerns parameter space. The sub-hypotheses sit in Carroll’s category 4. Beane et al. note that lattice improvement “masks much of our ability to probe” the scenario. Apart from dispersion-relation effects, $\mathcal{O}(b^2)$ operators easily avoid obvious probes [2]. A null result therefore excludes one simulator design at one resolution. A finer or improved lattice always survives. The same holds for naive preferred-frame discreteness whether or not it is simulated. Collins et al. show that it generically produces percent-level low-energy Lorentz violation unless it is strongly fine-tuned [99]. Constraints of this kind bear on the physics of discreteness. They bear on simulation only through the assumed design.

Table 2: Restricted simulation sub-hypotheses, their proposed signatures, and non-simulation physics that predicts the same signature. The status column is a summary only; see the sections cited for details.

Sub-hypothesis	Signature that would count as evidence	Degenerate non-simulation explanation	Status (summary)
H-grid: fixed lattice	Lattice-specific dispersion and energy cut-off; cubic-symmetric anisotropy of ultra-high-energy cosmic rays [2]	Lorentz violation in quantum-gravity models [91]; minimal-length scenarios [92]; Lorentz-invariant causal-set discreteness, which also removes the anisotropy [93]	Unimproved-Wilson design bounded to $b^{-1} \gtrsim 10^{16}$ GeV (this work); no cubic-anisotropy search published; see Secs. 5, 6
H-noise: finite resolution	Correlated Planckian position noise in co-located interferometers	Planckian quantum geometry with holographic degrees-of-freedom counting [94]	One model excluded at 4.6σ incl. calibration uncertainty [4]; further shear-noise limits [95]; see Sec. 7
H-lazy: render on demand	Interference outcomes that depend on the availability of which-path information to an observer [5]	Collapse models (GRW/CSL) [96]; von Neumann–Wigner observer postulate, as noted by the proposers [5]	Proposed schemes imply deviations from QM; none performed; related experiments agree with QM; see Sec. 9
H-budget: bounded compute	Physics that a bounded simulator cannot compute exactly is absent or approximated	Intractability as a physical principle [97]; physical limits on computation [52]	In-principle arguments, no empirical test; see Sec. 8
H-info: information-physical	New information-thermodynamic laws or effects	Information physics within standard physics [52]	Mass-information conjecture untested; reported entropy trends have standard explanations; see Sec. 10
H-patch: adaptive simulator	Glitches; slow drift of constants [43]	Varying-constant theories with a light field coupled to matter [98]	Constancy tests exist but are not simulation tests [98]

4.4 Adaptive simulators and the meaning of null results

The strongest obstacle to testing is a simulator that responds to being probed. Bostrom describes this case explicitly. A simulator tracking observers’ belief states could fill in microscopic detail “on an as-needed basis”. If an error occurred, it could “edit the states of any brains that have become aware of an anomaly” or “skip back a few seconds and rerun the simulation” [1]. Barrow argues that simulators with incomplete knowledge of physics would have to “patch up the problems one by one”. Inhabitants would then see occasional glitches and slow drifts in the constants [43]. Campbell et al. model a rendering engine that must preserve consistency while avoiding detection. They propose experiments intended to exploit the tension between these two requirements [5]. Hossenfelder questions whether adaptation is coherent. She asks how a programmer could notice that a simulated mind is about to detect a contradiction, and repair it in time, without predicting in advance what the mind will investigate [100].

Methodologically, adaptivity acts as an auxiliary hypothesis of the holist kind [81]. Suppose the simulator suppresses detectable anomalies. Then $P(\text{null} \mid H_{\text{SIM}} \wedge \text{adaptive}) \approx P(\text{null} \mid \neg H_{\text{SIM}})$, and a null result is approximately relevance-neutral [82]. Invoking adaptivity after a null result has the structure that Popper classed as an ad hoc rescue [80]. Rovelli’s point applies here too. A flexible theory loses Bayesian credibility with each failed prediction only when a success would have raised it [87]. For a fully adaptive simulator no outcome was expected, so null results cost it

nothing. Barrow’s patching scenario does yield a signature, drifting constants. That signature is itself degenerate with varying-constant physics, in which a varying constant reflects a light field coupled to matter [98].

We therefore state every null result in this review as conditional. It constrains simulators that are non-adaptive (they do not intervene in response to probes) and that implement the assumed architecture. The finite-resource premise does real work here. Beane et al. argue that finite simulator resources imply a non-zero lattice spacing, so that “in principle there always remains the possibility for the simulated to discover the simulators” [2]. That conclusion holds only under the finite-resource premise.

4.5 Published assessments of scientific status

Published assessments range from outright rejection of the hypothesis’s scientific standing to the view that it cannot be ruled out. Hossenfelder calls the hypothesis unscientific, “not a serious scientific argument”, while stating that this “doesn’t mean it’s wrong” [101]. Her central objection is that it assumes, without explanation, that all observations can be reproduced by an algorithm other than the confirmed laws. She adds that algorithmic reproductions of natural laws are usually incompatible with relativistic symmetries [101]. Aaronson argues that the Ringel–Kovrizhin result (Sec. 4.6) could not rule out the hypothesis. Among his reasons are that the simulator could be a quantum computer, or a classical one given exponentially more time [102]. Vazza derives energy requirements that he finds incompatible with physics for any simulator in a universe like ours [103]. He concludes that the hypothesis can be “reasonably well tested only with respect to universes which are at least playing according to the Physics play book”. Anything else, he writes, lies “beyond the bounds of falsifiability” [103]. Wolpert argues that experimental approaches can give only partial answers, because they rest on assumptions such as a digital simulating computer [89]. Chalmers argues that we should take the hypothesis seriously and cannot rule it out [22]. His epistemic asymmetry is discussed above (see also Sec. 2). The hypothesis was the subject of the 2016 Isaac Asimov Memorial Debate at the American Museum of Natural History, with physicists and a philosopher on the panel [104]. We cite the debate as an event only.

Among peer-reviewed work, the explicit discussions of testability that we found are physics or computer-science papers, not philosophy of science [2, 5, 89, 103]. We are not aware of a peer-reviewed philosophy-of-science article devoted specifically to the falsifiability of the simulation hypothesis. Our search is described in the accompanying research notes.

4.6 Reception: what was reported versus what was claimed

Table 3 pairs four results with how they were publicly reported. The news items and press releases cited here are evidence of reporting only, not of any scientific claim.

The four cases differ in kind. In the Beane et al. case the release is broadly faithful. It nevertheless elides the dependence on one assumed design, which the paper itself stresses [2, 105]. In the Ringel–Kovrizhin case the paper concerns a class of classical algorithms and never mentions the simulation hypothesis [6]. The simulation conclusion was added in reporting [106]. Aaronson responded publicly that the hypothesis “has not been falsified; remains unfalsifiable” [102]. One outlet reported that the authors were surprised by the headlines [110]. In the Vopson case the paper’s abstract is itself hedged (“appears to underpin”) [8]. The author’s statement, quoted in the coverage, that the proposed law holds whether or not the universe is simulated amounts to a concession of degeneracy [107]. In the Faizal et al. case the strong conclusion appears in the paper itself: the simulation hypothesis is “logically impossible rather than merely implausible” [7]. The

Table 3: Claims made in four papers compared with how the results were reported. News and press sources document reporting only.

Paper	What the paper claims	How it was reported
Beane et al. [2]	Bounds and a proposed signature for one assumed design (unimproved Wilson action, cubic lattice); improvement “masks” the signal	University release: “Researchers say idea can be tested”; “the first testable signature of such an idea” [105]
Ringel & Kovrizhin [6]	Quantized gravitational responses obstruct local sign-free quantum Monte Carlo; no mention of the universe or the simulation hypothesis	“Physicists Confirm That We’re Not Living In a Computer Simulation” [106]
Vopson [8]	Abstract: “scientific evidence that appears to underpin the simulated universe hypothesis”	“... Could Mean We Really Live in a Simulation”; author quoted: the law “is valid regardless of whether the universe is a simulation or not” [107]
Faizal et al. [7]	Incompleteness theorems applied to quantum gravity modelled as an axiomatic, algorithmic structure: “the universe cannot be a simulation”	“mathematically proven” and “a definitive answer” [108]; “Physicists prove the Universe isn’t a simulation after all” [109]

press release then presented it as a definitive mathematical proof [108]. The argument’s validity is disputed. A comment posted to arXiv argues that undecidability limits provability, not execution [111]. We assess this in Sec. 8. In every case the headline tied the result to the unrestricted hypothesis (“the idea”, “we”, “the Universe”). By the analysis of Sec. 4.2, no single result can settle that hypothesis.

4.7 Working thesis of this review

On this analysis we adopt the following position. The unrestricted simulation hypothesis is not an empirical hypothesis. No observation can refute it, and strong evidence for it would most plausibly require disclosure by the simulator itself [88]. What can be tested are conjunctions of the hypothesis with explicit assumptions about simulator architecture, resources and non-adaptivity. For each such sub-hypothesis in Table 2, the signature is also predicted by non-simulation physics. A detection would therefore establish new physics, such as discreteness, holographic noise or collapse. It would not establish that the universe is simulated. A null result bounds the space of non-adaptive, resource-limited simulator designs that share the assumed architecture. Accordingly, in what follows we speak of constraining simulator designs, and we avoid “proving” or “disproving” the simulation hypothesis.

5 A discretized spacetime as a simulation signature: theory and constraints

The most concrete physical proposal for testing the simulation hypothesis is that a simulator would discretize spacetime on a lattice, as lattice gauge theory does, and that the discretization would leave observable artefacts. This section covers four things:

- what the proposal assumes and predicts (Sec. 5.1);
- how lattice artefacts depend on the discretization (Sec. 5.2);
- which Lorentz-invariance-violation (LIV) bounds constrain it (Secs. 5.3–5.4);

- why these bounds exclude only simulators with a preferred frame (Secs. 5.5–5.6).

Section 5.7 lists what remains open. Cosmic-ray observations are reviewed in Sec. 6. Here we use only the energy scales. Results derived for this review are labelled *this work*. Their derivations are given in the supplementary material.

5.1 The lattice-simulation proposal of Beane, Davoudi and Savage

Assumptions. Beane, Davoudi and Savage [2] study “the hypothesis that the observed universe is a numerical simulation performed on a cubic space-time lattice”. They motivate it by extrapolating the resource requirements of lattice QCD, which scale roughly as $\lambda_{\text{QCD}} L^5/b^6$ for a lattice of extent L and spacing b . The extrapolation is fitted to the MILC (staggered) and SPECTRUM (anisotropic clover-Wilson) ensemble programmes. The authors list its caveats themselves, starting with the continuation of an effective Moore’s law [2, Sec. I]. They further assume:

- a classical computer;
- a hypercubic grid;
- following the history of lattice QCD, an “early” simulation with the cheapest discretization, unimproved Wilson fermions, whose leading artefacts are $O(b)$ [2, Secs. I–II].

In their analysis, all particles, including composite hadrons, obey the free lattice dispersion relations of their eqs. (16)–(17) [2, Sec. IV].

Derived constraints. Three results follow.

1. The $O(b)$ Pauli term of the Wilson action shifts magnetic moments by $2mbC_p$, with $C_p = 1 + O(\alpha)$ [2, eqs. (3), (5)]. Attributing the then-current $\sim 3.6\sigma$ muon $g - 2$ discrepancy entirely to the lattice gives $b^{-1} = (3.6 \pm 1.1) \times 10^7$ GeV, which the authors call “an approximate upper bound on the lattice spacing” [2, eq. (10)]. This is a central value tied to a 2012 anomaly, not a null bound.
2. Comparing α from the electron $g - 2$ with α from atomic recoil gives $b = |(-0.6 \pm 1.7) \times 10^{-9}|$ GeV $^{-1}$, hence $b^{-1} \gtrsim 4 \times 10^8$ GeV from the 1σ values [2, eqs. (12)–(13)].
3. Any lattice imposes a maximum energy $E_{\text{max}} \sim 1/b$. Requiring the cosmic-ray spectrum to extend to its observed cutoff gives $b^{-1} \sim 10^{11}$ GeV, the paper’s headline bound [2, Sec. IV].

Two qualifications apply. The authors quote the photopion (“GKZ”) cutoff as $\sim 6 \times 10^{20}$ eV. The conventional threshold is $\simeq 5 \times 10^{19}$ eV [112], and the Pierre Auger Observatory measures a flux suppression above $E_{34} = (46 \pm 3 \pm 6) \times 10^{18}$ eV [113]. The order of magnitude of the bound is unaffected, but it carries no confidence level. The authors also note that for chirally symmetric discretizations C_p vanishes or is exponentially small, so that the $g - 2$ and α constraints become much weaker. They regard all of these constraints as “estimates only” [2, Sec. III].

The predicted signature. The free lattice dispersion relations have only cubic symmetry [2, eqs. (16)–(17), Fig. 2]. If the lattice rest frame coincides with the CMB frame, the Δ -resonance threshold for photopion production becomes direction dependent [2, eq. (18)]. Rewriting that equation (this work), the fractional threshold shift is

$$\frac{\delta\omega}{\omega} = \frac{5}{12} \left(\sum_j n_j^4 - \frac{3}{5} \right) (b|\mathbf{p}|)^2, \quad (3)$$

where $\mathbf{n} = \mathbf{p}/|\mathbf{p}|$ is taken in the lattice frame. The shift is $+(b|\mathbf{p}|)^2/6$ along a lattice axis and $-(b|\mathbf{p}|)^2/9$ along a body diagonal.

Beane et al. predict that, if the lattice supplies the cutoff, “the angular distribution of the highest energy components would exhibit cubic symmetry in the rest frame of the lattice”. For smaller spacings the effect would be less significant and masked by the GZK mechanism, and a non-cubic lattice would imprint its own symmetry [2, Sec. IV]. The paper gives no arrival-direction amplitude beyond eq. (18). It does not model magnetic deflection or the source distribution. Among the physics papers that cite Ref. [2], we found one direct technical response [114] and one that uses the UHECR energy scale in an energy-cost argument [103]. None re-derives the bounds or tests the anisotropy against data (see the notes for the citation search).

Two further statements matter below. First, the authors argue that the naturalness problem of Lorentz-violating dimension-four operators does not arise, because “the underlying space-time symmetries respected by the lattice action will necessarily be preserved at the quantum level”. They attribute this protection to Euclidean hypercubic symmetry and note that Hamiltonian formulations are also possible [2, Sec. I and fn. 3]. Second, they conclude that improvement “masks much of our ability to probe” the scenario, and that it is “unlikely that any but the very earliest universe simulations would be unimproved” [2, Sec. V].

5.2 Lattice artefacts depend on the discretization

The predictions above are properties of one action, not of lattices in general.

- **Fermion discretizations.** The Nielsen–Ninomiya theorem forbids a local lattice fermion that is free of doublers, chirally symmetric and has the correct continuum limit [115]. Every fermion discretization therefore trades one property for another. Wilson fermions generically have $O(a)$ artefacts. Staggered fermions, twisted-mass fermions at maximal twist and exactly chiral (domain-wall or overlap) fermions scale as $O(a^2)$. Improved staggered actions such as asqtad scale as $O(g^2 a^2)$ [115].
- **Gauge action.** The plaquette action has $O(a^2)$ errors. Tree-level Lüscher–Weisz improvement removes them classically, but radiative corrections reintroduce $O(g^2 a^2)$ terms [115].
- **Rotational symmetry.** On a hypercubic lattice the action is constrained by hypercubic symmetry. Davoudi and Savage show that, with suitably smeared operators, rotation-violating contributions are suppressed by $O(a^2)$ in the continuum limit, including at one loop [116].
- **Anisotropic lattices.** These have temporal spacing a_t different from spatial spacing a_s (e.g. $a_s/a_t = 3.5$). The gauge and fermion anisotropies must be tuned so that boosted hadrons obey a relativistic dispersion with a common “speed of light” [117].
- **Symmetric alternatives.** Discretizations that retain continuous symmetry have been proposed. A Euclidean regulator with a dynamical embedding is exactly $SE(d)$ covariant at finite spacing [118]. A “Lorentz covariant” lattice-graph construction has been proposed explicitly in reply to Beane et al. [114]; it is an unrefereed preprint whose technical claims we have not assessed.

Any bound on b is therefore a bound on a particular simulator design.

Table 4: Representative LIV bounds relevant to a spacetime lattice. $E_{\text{Pl}} = 1.22 \times 10^{19}$ GeV. “Sub” means subluminal. n follows the convention of each source.

Observable	Bound	CL	Ref.
GRB 090510 time of flight (Fermi)	$M_{\text{QG},1}/M_{\text{Pl}} > 1.19$ (most conservative) to > 102 ; > 1.22 from $ \Delta t/\Delta E < 30$ ms/GeV	99% for the lag analysis	[125]
Four Fermi-LAT GRBs	$E_{\text{QG},1} > 7.6 E_{\text{Pl}}$, $E_{\text{QG},2} > 1.3 \times 10^{11}$ GeV (sub, no intrinsic effects); $1.8 E_{\text{Pl}}$ and 4.0×10^{10} GeV with intrinsic effects	95%	[126]
GRB 221009A (LHAASO)	$E_{\text{QG},1} > 1.0 \times 10^{20}$ GeV, $E_{\text{QG},2} > 6.9 \times 10^{11}$ GeV (sub); abstract: “ $> 10 E_{\text{Pl}}$ ”, “ $> 6 \times 10^{-8} E_{\text{Pl}}$ ”	95%	[127]
PeV photons, no spectral cutoff (LHAASO)	$E_{\text{cut}} > 1140$ TeV (J2032+4102), > 750 TeV (Crab); superluminal photon scale $\approx 1.42 \times 10^{33}$ eV ($n = 1$)	95%	[128]
Vacuum birefringence, GRB 140206A	$\xi < 1 \times 10^{-16}$ (CPT-odd, $n = 3$)	— ^a	[129]
Crab synchrotron, leptons, $n = 3$	$O(10^{-5})$ (subluminal max-velocity argument introduced in [130])	95%	[131]
Crab synchrotron, leptons, $n = 4$	$M_{\text{LV}} \gtrsim 2 \times 10^{16}$ GeV	95%	[9]
UHECR spectrum, protons, dim. 6	$-10^{-3} \lesssim \eta_p^{(4)} \lesssim 10^{-6}$ (pure protons)	99%	[112, 132]
GZK photons (Auger)	$\delta_{\gamma,2} > -10^{-58}$ eV ⁻² (only with subdominant protons to 10^{20} eV)	— ^b	[133]
Causal-set swerves (relic ν)	$k < 10^{-61}$ GeV ³	estimate	[134]

^a The limit follows from requiring polarization rotation $\leq 90^\circ$ across the band. The polarization itself is $> 28\%$ at 90% c.l.

^b Hadronic limits in the same work ($\delta_{\text{had},2} < 10^{-57}$ eV⁻² at 5σ) cover only positive, superluminal coefficients.

5.3 The LIV framework and current bounds

A lattice at rest in some frame breaks Lorentz invariance, so LIV constraints apply to it. The Standard-Model Extension (SME) parametrizes Lorentz and CPT violation in effective field theory [119, 120]. Its coefficients are compiled in the Kostelecký–Russell data tables, whose January 2026 edition (arXiv v19) contains 52 data tables covering the literature to the end of 2025 [121]. Coleman and Glashow’s earlier framework considers renormalizable, rotationally invariant perturbations in a preferred frame, which give species-dependent maximal attainable velocities [122]. Reviews treat the modified dispersion relations $E^2 = p^2 + m^2 + f^{(n)} p^n / E_{\text{Pl}}^{n-2}$ and their astrophysical tests [91, 112, 123, 124].

Two features matter here. First, the QED dimension-5 operators, which give $n = 3$ dispersion and vacuum birefringence, all violate CPT [123]. Second, once photon decay is kinematically allowed, the photon lifetime is “extremely short ($\ll 1$ sec)”. Threshold constraints are therefore insensitive to the dynamics [123]. Table 4 collects representative bounds.

5.4 Mapping the lattice dispersion onto LIV bounds

Beane et al. note correctly that the Coleman–Glashow velocity bounds “do not directly apply”, because lattice velocities depend on momentum [2, Sec. IV]. The momentum-dependent LIV literature

does apply, however.

The dispersion relations as dimension-six LIV (this work). Expanding eqs. (16)–(17) of Ref. [2] for $|\mathbf{k}| \ll 1/b$ gives

$$E_\gamma^2 = k^2 - \frac{b^2}{12} f(\mathbf{n}) k^4, \quad E_e^2 = k^2 + m^2 - \frac{b^2}{3} f(\mathbf{n}) k^4, \quad f(\mathbf{n}) = 1 + \sum_j n_j^4 \in [\frac{4}{3}, 2]. \quad (4)$$

We checked these coefficients against the exact relations numerically. For a continuous-time (Hamiltonian) lattice, $f \rightarrow \sum_j n_j^4 \in [\frac{1}{3}, 1]$. The modification is even in k and has a single photon branch. The scenario is therefore CPT-even and non-birefringent at leading order, and birefringence limits [129, 135] do not constrain it.

Consequences (this work). Four constraints follow. We quote each at the least constraining lattice orientation.

1. *Time of flight.* Matching Eq. (4) to the LHAASO parametrization gives $E_{\text{QG},2} = b^{-1}(12/f)^{1/2}$. The LHAASO limit [127] then implies $b^{-1} \gtrsim 2 \times 10^{11}$ GeV at 95% CL. This is comparable to the E_{max} estimate, but it comes from an independent observable with a stated confidence level.
2. *Photon decay.* In Eq. (4) the electron is four times more subluminal than the photon at equal momentum. Asymmetric pair production $\gamma \rightarrow e^+e^-$ is then kinematically allowed above $k_{\text{th}} = (64/f)^{1/4}(m_e/b)^{1/2}$, far below $1/b$. We confirmed this numerically with the exact lattice relations. The result differs from the statement in Ref. [2] that this process proceeds only for photon energies “comparable to the inverse lattice spacing”. Requiring no decay below the LHAASO lower limit on any cutoff, 1140 TeV [128], gives $b^{-1} \gtrsim 4 \times 10^{14}$ GeV. This assumes, as argued in the EFT literature [112, 123], that decay above threshold is fast. We have not computed the decay rate in the lattice theory itself.
3. *Crab synchrotron.* The lattice electron has an $n = 4$ subluminal dispersion with $\eta = -1$ and $M_{\text{LV}} = b^{-1}(3/f)^{1/2}$. The Crab-Nebula limit $M_{\text{LV}} \gtrsim 2 \times 10^{16}$ GeV (95% CL) [9] then implies $b^{-1} \gtrsim 10^{16}$ GeV.
4. *UHE photons.* With a sub-dominant proton component, the Auger photon limit [133] would imply $b^{-1} \gtrsim 3 \times 10^{19}$ GeV. This limit is conditional on the unsettled UHECR composition [112].

These mappings use EFT bounds derived for isotropic dispersion, so they hold only up to factors of order unity. Direction-dependent SME limits [121, 126] would be the rigorous route.

For the unimproved-Wilson scenario, the strongest robust constraint is therefore not the GZK argument but the $n = 4$ electron bound. By Eq. (3), it limits the cubic anisotropy of the photopion threshold near 5×10^{10} GeV to a fraction of order 10^{-11} .

Improved actions (this work). The answer depends on which bound is considered.

- The $E_{\text{max}} \sim 1/b$ argument depends only on the Brillouin zone, $|k_j| \leq \pi/b$. It changes by factors of order unity under improvement.
- The $g - 2$ and α bounds disappear once the Pauli term is removed [2].
- $O(b)$ (clover) improvement leaves the free dispersion, and hence all of the bounds above, unchanged.

- With improvement only at tree level, residual $O(ab^2)$ terms [115] would still give $b^{-1} \gtrsim 10^{15}$ GeV via the Crab bound.
- If the $O(b^2)$ terms are removed to all orders, the leading anisotropy becomes $O(b^4 k^4)$. The synchrotron maximum-frequency argument of Ref. [9], applied to $n = 6$, then gives only $b^{-1} \gtrsim 10^{11}$ GeV, comparable to the Beane et al. estimate.

A cubic-anisotropy signal at the GZK scale therefore requires a simulator that is both nonperturbatively improved and has b^{-1} within about an order of magnitude of 10^{11} GeV.

5.5 Naturalness: why preferred-frame discreteness is hard to hide

Collins et al. argue that a Planck-scale preferred frame, combined with known particle interactions, induces Lorentz violation “at the percent level, some 20 orders of magnitude higher than earlier estimates, unless the bare parameters of the theory are unnaturally strongly fine-tuned”. The violation appears as species-dependent speeds of light [99]. Their argument assumes real time rather than a Euclidean continuation.

Gambini et al. argued that non-perturbative treatments may evade this conclusion [136]. Polchinski replied that a Euclidean lattice with equal steps has discrete rotational symmetries that forbid dimension-four Lorentz-violating terms, whereas “a Lorentzian lattice with equal time and space steps has no such enhanced symmetry” and a lattice propagator is not consistent with unitarity. He concludes that the only known exceptions rely on supersymmetry [137]. Later work shows that a separation between the EFT scale and the LIV scale can suppress percolation to low energies [138].

This debate bears directly on the claim of Beane et al. that hypercubic symmetry protects the simulation. That protection is a property of Euclidean lattices, and a real-time or Hamiltonian simulator lacks it. The need to tune separate gauge and fermion “speeds of light” on anisotropic lattice-QCD ensembles [117] is a concrete instance of the dimension-four tuning that Collins et al. describe.

5.6 Lorentz-invariant discreteness

Discreteness does not require a preferred frame. In causal set theory [139], spacetime is a random Poisson sprinkling, and its discreteness is locally Lorentz invariant [93]. Bombelli, Henson and Sorkin proved that no equivariant measurable map exists from sprinklings to spacetime directions. The discreteness of a sprinkled causal set therefore “will not give rise to ‘Lorentz breaking’ effects like modified dispersion relations”. The same theorem implies that no finite-valency graph can be associated with a sprinkling consistently with Lorentz invariance [140]. In manifold-like causal sets each element has infinitely many nearest neighbours, and the resulting non-locality is characteristic of the approach [141]. On our reading, this non-locality is the computational price of a Lorentz-invariant discretization.

The null LIV results in Table 4 therefore exclude only simulators with a lattice at rest in some frame. They are silent about Lorentz-invariant discretizations of this type.

Causal sets have their own phenomenology: Lorentz-invariant momentum-space diffusion, or “swerves” [93, 142]. Relic-neutrino bounds give $k < 10^{-61}$ GeV³. With effective-field-theory scaling this requires a dimensionless coefficient below 10^{-118} , which Kaloper and Mattingly call “an extremely small value begging for a first-principles explanation” [134]. More generally, discreteness is neither necessary nor sufficient for a minimal length [92].

5.7 What remains open

1. We are not aware of any published search for cubic-symmetric anisotropy in UHECR arrival directions. Our searches of INSPIRE, arXiv and Semantic Scholar are documented in the notes; see also Sec. 6. The mapping above suggests that such a search is informative only for nonperturbatively improved simulators with $b^{-1} \sim 10^{11}$ GeV.
2. The mappings in Sec. 5.4 should be redone with direction-dependent SME coefficients and with the dispersion relations of domain-wall and overlap fermions, which we have not derived.
3. For Lorentz-invariant discreteness, two gaps remain. Quantum fields on causal sets have so far been constructed only for the free scalar [124]. Extending the swerve calculation to an FRW universe is open [141].
4. A lattice-like signature does not by itself distinguish a simulated universe from discrete physics. Every bound discussed here is a bound on a class of discretizations, whether or not anyone is running them.

6 Ultra-high-energy cosmic rays as a probe of a cubic spacetime lattice

This section covers the observational side of the proposal by Beane, Davoudi and Savage [2]. It reviews what measurements of ultra-high-energy cosmic rays (UHECRs) say about their lattice scenario, separates what current data constrain from what they leave open, and ends with a specific analysis that could be run on public data. The lattice theory and the Lorentz-violation bounds are treated in Sec. 5.

6.1 The prediction and its premises

Beane *et al.* assume a simulation on a hyper-cubic space-time grid with spacing b , starting from an unimproved Wilson action [2]. The lattice dispersion relations cap the energy at $E_{\max} \sim 1/b$ for fermions and bosons. Setting this cap equal to the Greisen–Zatsepin–Kuzmin (GZK) cutoff [143, 144] gives $b \sim 10^{-12}$ fm, that is $b^{-1} \sim 10^{11}$ GeV. The authors turn this into the bound $b^{-1} \gtrsim 10^{11}$ GeV, their most stringent one [2]. They also argue that if the lattice itself supplies the cutoff, “the angular distribution of the highest energy components would exhibit cubic symmetry in the rest frame of the lattice”, and that for smaller b the cubic signal would fade while the GZK mechanism takes over [2]. Their Eq. (18) supports this at leading order. There the lattice shifts the Δ -resonance threshold of proton–CMB-photon collisions in proportion to the combination $Y_4^0 + \sqrt{5/14}(Y_4^4 + Y_4^{-4})$, taking the lattice rest frame to be the CMB frame [2].

The prediction has three limits that matter here. (i) It is qualitative. The paper gives neither the size of the flux anisotropy as a function of b nor a sky template. (ii) It assumes that the observed steepening is the GZK cutoff, and it treats the primaries through single-particle dispersion relations. The paper does not discuss heavy nuclei as primaries [2]. We add that a nucleus of mass number A carries only about E/A per nucleon, so identifying $E_{\max}$ with the observed endpoint depends on composition. (iii) The arXiv version puts the GZK scale at “ $\sim 6 \times 10^{20}$ eV” [2]. That is ten times the threshold of $\sim 6 \times 10^{19}$ eV quoted by HiRes [145], and we do not reproduce it.

We checked numerically which spherical-harmonic degrees admit functions invariant under the full cubic group O_h (48 elements). Up to $\ell = 12$ they are $\ell = 0, 4, 6, 8, 10$ and 12 (two at $\ell = 12$). The combination in Beane *et al.*’s Eq. (18) is the $\ell = 4$ invariant, and $Y_6^0 - \sqrt{7/2}(Y_6^4 + Y_6^{-4})$ is the

$\ell = 6$ one. It follows that a lattice-aligned pattern has no dipole and no quadrupole: its leading content is at $\ell = 4$ and $\ell = 6$.

6.2 The flux suppression and the composition question

HiRes reported a suppression at $\log_{10}(E/\text{eV}) = 19.75 \pm 0.04$. It observed 13 events above $10^{19.8}$ eV where an unbroken spectrum predicted 43.2, a Poisson deficit equivalent to 5.3 standard deviations [145]. The Pierre Auger Observatory rejected a constant slope above 4×10^{19} eV at 6 standard deviations, while cautioning that identifying the cause would require knowledge of the mass composition [146]. Auger’s 2020 spectrum uses 215,030 events and an exposure of $(60,400 \pm 1,810) \text{ km}^2 \text{ sr yr}$ [147]. It places the suppression at $E_{34} = (46 \pm 3 \pm 6) \times 10^{18}$ eV with spectral index $\gamma_4 = 5.1 \pm 0.3 \pm 0.1$, and finds a new softening (the “instep”) at $E_{23} = (13 \pm 1 \pm 2) \times 10^{18}$ eV [113]. The measured $E_{1/2} = (2.2 \pm 0.1 \pm 0.3) \times 10^{19}$ eV differs from the 5.3×10^{19} eV expected for uniformly distributed, proton-only sources, a scenario Auger describes as disfavored [113].

Mass composition decides how the endpoint should be read. In Auger’s benchmark mixed-composition model, the steepening above $\approx 5 \times 10^{19}$ eV comes from the maximum acceleration energy of the heaviest nuclei combined with the GZK effect. Auger adds that a subdominant light component is not excluded and that other scenarios remain viable [113]. A combined fit of spectrum and composition prefers sources with a mixed composition, a hard spectrum and a low rigidity cutoff [148]. Auger has also inferred X_{max} event by event from surface-detector data up to 100 EeV [149]. That analysis finds a composition that becomes “increasingly heavier and purer” and is incompatible with a large light fraction between 50 and 100 EeV. It concludes that the suppression “cannot be entirely ascribed to effects of extragalactic propagation” [149]. Telescope Array (TA) hybrid data are compatible with QGSJet II-04 protons at 95% CL in all energy bins, after systematic shifts. At the highest energies, where statistics are sparse, other elements are also compatible [150]. In short, the premise that the observed endpoint is the GZK cutoff of a proton flux is disputed by current data. This weakens the inference from “the spectrum ends near 10^{20} eV” to a statement about b , whether the end is attributed to the GZK effect or to a lattice.

6.3 Arrival directions at each angular scale

Dipole. Auger observed a dipole above 8 EeV at more than 5.2σ (post-trial), using 3×10^4 events from an exposure of $76,800 \text{ km}^2 \text{ sr yr}$. Its amplitude is $6.5_{-0.9}^{+1.3}\%$ towards $(\alpha, \delta) = (100 \pm 10^\circ, -24_{-13}^{+12}^\circ)$ [151]. With 19 years of data (49,678 events ≥ 8 EeV), the right-ascension modulation reaches 6.8σ . The 3D dipole is $7.4_{-0.8}^{+1.0}\%$ towards $(97 \pm 8^\circ, -38 \pm 9^\circ)$, and it does not vary in time: the rate of change is below 0.3% per year at 95% CL [152]. In the joint dipole-plus-quadrupole fit, the quadrupole amplitude for $E \geq 32$ EeV is $Q = 0.13 \pm 0.06$, and the quadrupolar components are not significant [152].

Intermediate scales. A starburst-galaxy sky model fits 5,514 Auger events above 20 EeV better than isotropy at 4.0σ [153]. With the full Phase 1 set of 2,635 events above 32 EeV (exposure $122,000 \text{ km}^2 \text{ sr yr}$), Auger finds evidence at the 4σ level for an excess on a $\sim 15^\circ$ (Gaussian) to $\sim 25^\circ$ (top-hat) scale above ~ 40 EeV. The most significant blind excess lies in the Centaurus region, with 5.4σ local significance and a 3% post-trial probability [154]. In the north, TA reported a hotspot above 57 EeV with local significance 5.1σ and chance probability 3.7×10^{-4} (3.4σ) [155]. The joint Auger–TA update finds the starburst correlation to be the most significant, at 4.2σ post-trial [156]. TA also recorded a single event of 244 ± 29 (stat) $_{-76}^{+51}$ (syst) EeV whose direction points back to a void [157].

Table 5: Published analyses that measured UHECR multipoles beyond $\ell = 2$, all of which therefore contain $\ell = 4$ and $\ell = 6$. None forms an O_h -invariant combination or scans lattice orientations.

Analysis	Data, energy	ℓ range	Result beyond the dipole
Auger 2017 [158]	Auger, 4–8 and ≥ 8 EeV	$C_\ell, \ell \leq 64$	no deviation from isotropy
Auger 2024 [152]	Auger, 4 bins ≥ 4 EeV	$C_\ell, \ell \leq 20$	$\hat{C}_{17}, \hat{C}_8$ not significant after penalty; 99% UL graphical
Auger+TA 2014 [159]	full sky, > 10 EeV	$a_{\ell m}, C_\ell, \ell \leq 20$	no significant deviation
Auger+TA 2025 [156]	full sky, threshold scan	C_ℓ (auto, cross), $\ell \leq 20$	$\ell = 2$ most significant (2.1σ post-trial)
Ahlers 2018 [160]	public Auger, > 8 EeV	pseudo- C_ℓ	consistent with isotropy

Higher multipoles, including $\ell = 4$ and $\ell = 6$. Table 5 lists the analyses that measured multipoles beyond the quadrupole. Auger observes only part of the sky, so it cannot estimate individual $a_{\ell m}$ with useful resolution once $\ell_{\max} > 2$. It reconstructs C_ℓ instead, assuming the anisotropies are one realization of a statistically isotropic random process [152, 158]. The 2024 spectrum (with $\ell = 1$ –20 in each of four energy bins) shows no significant $\hat{C}_\ell$ apart from the dipole. Two excursions, $\hat{C}_{17}$ at 4–8 EeV and $\hat{C}_8$ at 16–32 EeV, have penalized chance probabilities of 3.3% and 26.5% [152]. The 99% CL upper limits on C_ℓ are shown only graphically; no numerical C_4 or C_6 limits are tabulated [152]. With full-sky Auger+TA data, the $a_{\ell m}$ have been estimated up to $\ell = 20$ above 10^{19} eV, with no evidence for anisotropy [159]. A newer harmonic-space analysis over $\ell \leq 20$ with a scan in energy threshold finds the quadrupole to be the most significant multipole in every case. Its largest auto-correlation excursion is at $\ell = 2$ above 41 EeV (Auger energy scale), at 4.2σ pre-trial and 2.1σ post-trial [156]. An independent likelihood reanalysis of the public 2017 Auger events likewise finds nothing beyond the dipole [160].

6.4 Magnetic smearing

Deflections scale with charge and with the integrated transverse field, and inversely with energy [151]. At $E \approx 10$ EeV, estimates of the mean charge range from $Z = 1.7$ to 5, and typical Galactic deflections are a few tens of degrees at $E/Z = 10$ EeV [151]. In the Jansson–Farrar (JF12) model, a 60 EeV proton is deflected by 5.2° on average. A quarter of the sky has deflections below 2.2° , and deflections in the south are about 60% larger than in the north [161]. The UF23 ensemble of eight coherent-field models [162] requires a rigidity of $R \geq 20$ EV for deflections below 20° over half the sky in all models. If directions are corrected for the modelled field, the requirement falls to 11 EV; the authors call this estimate indicative because random fields are not included [162]. The model-to-model spread is smaller than the deflections themselves over most of the sky [162].

For iron ($Z = 26$) at the suppression energy $E_{34} \approx 46$ EeV [113], the rigidity is only about 1.8 EV, far below these thresholds. The characteristic angular scale of multipole ℓ is about $180^\circ/\ell$ [152], or about 45° for $\ell = 4$ and 30° for $\ell = 6$. Unless the highest-energy particles are light, the Galactic field would therefore distort a lattice-aligned pattern in a direction-dependent way. A search must forward-model this distortion, or use the high-rigidity limit as the unperturbed case.

6.5 What the existing data do and do not constrain

C_ℓ does not change under rotations. A cubic pattern of *any* orientation therefore adds power to C_4 and C_6 , and the non-detections in Table 5 already bound the size of such a pattern without any orientation scan. The bound is only implicit, for four reasons. (a) No paper quotes numerical C_4 or

C_6 limits above the suppression energy. (b) The partial-sky C_ℓ reconstruction assumes statistical isotropy, which a fixed lattice-aligned sky violates [158]. (c) C_4 and C_6 each spread their statistical weight over $2\ell + 1 = 9$ and 13 directions and ignore the fixed phase relation that a cubic pattern imposes between $\ell = 4$ and $\ell = 6$. Blind scans over many multipoles lose sensitivity: for a 5% dipole in 14,000 simulated events, a blind $\ell \leq 64$ power-spectrum search detects it at 99% CL in only 25% of skies at $\delta = -30^\circ$ [158]. (d) Turning an amplitude limit into a limit on b requires the anisotropy amplitude as a function of b and rigidity, which Beane *et al.* do not provide [2]. Section 5 argues further that, for the unimproved Wilson action, existing Lorentz-violation bounds already suppress the cubic anisotropy of the photopion threshold. A detectable signal would then require an improved discretization with b^{-1} near 10^{11} GeV. These considerations limit what a positive or null result could mean. They do not make the search uninformative about the phenomenology.

Novelty check. We searched for any analysis that tests UHECR (or other astroparticle) arrival directions for a lattice-aligned, cubic or O_h -symmetric anisotropy, or that confronts the “numerical simulation” hypothesis with such data. We checked every work that cites Ref. [2] in INSPIRE (13 records) and Semantic Scholar (61 records), ran full-text queries in INSPIRE, and ran arXiv and general web searches. The full log is in the accompanying research notes. We found no such analysis. Citing works are theory, philosophy or popular accounts. Among them, Ref. [103] uses the UHECR energy only as a resolution requirement for a simulator, and Ref. [163] bounds a cubic-lattice quantum cellular automaton with photon timing and laboratory data. The closest data analyses test *direction-dependent Lorentz violation*, not a cubic flux pattern: vacuum-Cherenkov bounds on anisotropic modified-Maxwell coefficients from the directions of UHECR events [164], and a sidereal-modulation search in IceCube atmospheric neutrinos [165]. Auger’s own Lorentz-violation study uses direction-independent dispersion relations [133]. To our knowledge, no search for an O_h -symmetric UHECR anisotropy has been published. NASA ADS and Google Scholar could not be queried automatically, so this statement carries that caveat.

6.6 Public data

Auger opened its Open Data Portal in February 2021 with 10% of Phase I data. The release now holds 81,121 showers, including 25,086 events from the 1500 m array: vertical above 2.5 EeV and inclined (60° – 80°) above 4 EeV. The data come as JSON files per event and a CSV summary [166, 167]. In June 2023 the collaboration approved raising the public fraction to 30% of main-array events above 2.5 EeV from 2004–2022 (about $24,000 \text{ km}^2 \text{ sr yr}$), with release planned for late 2025 [168]. When we checked the portal and Zenodo on 27 September 2026, the current version was still the 10% release of March 2024 [167]. Our count of the public summary files gives only 86 events above 40 EeV. Much more useful for this problem is the *complete* Phase 1 list of 2,635 events above 32 EeV. It was published with Ref. [154] and gives, for each event, its time, local and equatorial coordinates, energy and cumulative exposure [169]. Our count of that file gives 1,387 events at $E \geq 40$ EeV and 411 at $E \geq 57$ EeV. Northern coverage is thinner in public form: TA published the 72 events above 57 EeV from its first five years [155], and Auger catalogued its 100 highest-energy events (78–166 EeV) [170]. An external all-scale reanalysis of public Auger events has already been published [160], so such work is practical.

6.7 A proposed search (not a result)

The following analysis is feasible with the public data above and addresses points (b) and (c) of Sec. 6.5. We present it as a proposal; it has not been carried out.

Model. The relative flux is written as $\Phi(\hat{n}) \propto \omega(\hat{n}) [1 + \vec{d} \cdot \hat{n} + \epsilon K(R^{-1}\hat{n}) + \dots]$. Here ω is the directional exposure, $\vec{d}$ is a nuisance dipole (plus quadrupole) with priors from Refs. [151, 152], and $K = \cos \chi \hat{K}_4 + \sin \chi \hat{K}_6$ is a unit-normalized combination of the $\ell = 4$ and $\ell = 6$ O_h invariants. The rotation $R \in SO(3)/O$ fixes the lattice orientation; the quotient is by the 24 proper rotations of the cube, since even- ℓ functions are automatically inversion-symmetric.

Statistic. The test statistic is the likelihood ratio $T = \max_{R, \chi} 2 \ln[\mathcal{L}(\epsilon, R, \chi, \vec{d})/\mathcal{L}(0, \vec{d})]$ from an unbinned likelihood over events. For a template-free cross-check, the “cubicness” of the measured $\ell = 4$ multipole, $\max_R |\langle \hat{K}_4, D^{(4)}(R) a_4 \rangle|^2 / \sum_m |a_{4m}|^2$, equals 1 for an exactly cubic $\ell = 4$ component and has a computable null distribution. The two degrees are tied together by sharing a single R , which is what distinguishes the test from separate C_4 and C_6 measurements.

Look-elsewhere correction. Isotropic-plus-dipole mock skies drawn with the actual exposure give the null distribution of T . This distribution already includes the three-parameter orientation scan and the χ scan. Energy thresholds (for example 32, 40 and 57 EeV on the Auger scale) should be fixed in advance, or else penalized in the same simulations.

Exposure and coverage. With Auger data alone, the unobserved northern cap correlates the $a_{\ell m}$. The likelihood in pixel or event space handles this directly and avoids the statistical-isotropy assumption behind pseudo- C_ℓ deconvolution. Adding the TA list [155] needs a cross-calibration of energy scales of the kind used in the joint analyses [156, 159].

Magnetic forward model. Lattice templates defined outside the Galaxy should be propagated through each of the eight UF23 variants [162] and JF12 [161] for a grid of rigidities. The spread of results is a systematic uncertainty, and the unperturbed template is the high-rigidity limit.

Validation. Before touching the data, one should inject cubic patterns of known (ϵ, R, χ) into mock skies that also contain the measured dipole and a Centaurus-like hotspot [154]. The checks are that the orientation is recovered, that the false-positive rate is correct, and what the detection power is as a function of ϵ . An idealized estimate that we verified by Monte Carlo sets the scale: for N events distributed isotropically with uniform full-sky exposure, the amplitude estimator of one unit-normalized harmonic has standard deviation $\sqrt{4\pi/N}$, about 0.095 for $N = 1,387$. Partial coverage, the nuisance dipole and the orientation scan will increase the threshold for detection.

Interpretation. A null result would bound ϵ as a function of energy threshold and assumed rigidity. Relating ϵ to b needs a calculation of the anisotropy amplitude that the literature does not yet contain (Sec. 5). The bound also does not apply to discretizations with other symmetries or with improved actions, which Beane *et al.* themselves consider likely [2].

7 Finite information density and interferometric searches for Planckian noise

A simulation run on finite hardware can only represent a finite amount of information about any region. Holographic entropy bounds say that, if semiclassical gravity is a reliable guide, nature also stores only a finite amount of information in a region, and that this amount scales with the area of the boundary rather than with the volume. This section reviews what those bounds imply about the “resolution budget” of any substrate that reproduces our physics. It then turns to the one experimental programme that has tried to detect a macroscopic consequence of finite spatial information: interferometric searches for Planckian “holographic noise”. The models tested in these searches come from quantum gravity, not from assumptions about simulation. We state this at the start because the popular description of these experiments as tests of whether space is “pixelated” blurs the distinction.

7.1 Holographic bounds as an information budget

Bekenstein argued from the generalized second law (GSL) that a weakly gravitating system of energy E and radius R in asymptotically flat space obeys $S_{\text{matter}} \leq 2\pi ER$ in Planck units [171]; we rely on Bousso’s account of the argument, which also notes that whether the GSL implies this bound “remains controversial” [172]. ’t Hooft went further and argued that the observable degrees of freedom can be described as Boolean variables on a two-dimensional lattice, about “one Boolean variable per Planckian surface element” [173]. Susskind recast this proposal as a “screen” of “pixels”, each storing one bit [174]. That is the origin of the pixel vocabulary, and it was introduced as a heuristic, not as a claim about computation. In Bousso’s review the covariant entropy bound limits the information content of the spacetime regions adjacent to a surface to 1.4×10^{69} bits per square metre, and the holographic principle is the assertion that this bound has its origin in a quantum theory of gravity [172]. Bousso also spells out the contrast with a naive lattice. One oscillator with n states per Planck volume gives $\sim n^V$ states, which scales with volume and is “at odds” with the spherical entropy bound [172].

These results bear on the simulation hypothesis in three ways. First, they give a physically motivated upper bound on the memory a faithful simulation of a region would need: a substrate that reproduces semiclassical gravity need not store more than an area-law number of bits. For the whole observable universe, Lloyd estimated at most $\sim 10^{120}$ elementary operations on $\sim 10^{90}$ bits ($\sim 10^{120}$ bits if gravitational degrees of freedom are included) [52]. Vazza used the holographic bound to estimate $\sim 3.5 \times 10^{124}$ bits and concluded that a Planck-resolution simulation of the visible universe by a simulator sharing our physics is impossible on energy grounds [103]. Vazza also notes that a coarse-grained description of the cosmic web needs vastly fewer bits, so the conclusion depends on the resolution assumed [103]. The resource arguments themselves are reviewed in the section on computational limits. Second, the bounds limit *maximum entropy*. They do not imply that geometry is discrete, or that a finite information density must leave a measurable imprint. ’t Hooft himself remarked that the holographic “blurring is small compared to the uncertainties produced by the usual quantum mechanical fluctuations” [173]. Third, whether the information limit could have *macroscopic* observable consequences is the question that motivated the experiments below.

7.2 Hogan’s holographic noise and the GEO600 episode

Hogan proposed that a transverse indeterminacy of position, inferred from the limits of measurement with Planck-wavelength radiation, would produce “holographic noise”. This noise would have a transverse shear signature and a flat strain power spectral density equal to the Planck time t_P , predicted “with no parameters” [3]. The step from information to noise is explicit. The normalization uses Nyquist–Shannon sampling with one degree of freedom per $2l_P$ in each transverse direction on a light sheet, and an experimental upper bound on the noise would therefore give “a lower bound on the number density of independent position eigenstates” [3]. Hogan stated that the effect is “not derived here from a fundamental theory”. He compared the predicted spectrum with GEO600, not LIGO, arguing that LIGO’s arm cavities suppress the effect [3], and he wrote that the approximate agreement with “otherwise unexplained noise in GEO600 motivates further study” [3, 175]. A 2009 revision lowered the predicted amplitude by $\sqrt{\pi}$, to $h = \sqrt{t_P/2\pi} = 0.92 \times 10^{-22} \text{ Hz}^{-1/2}$ for GEO600. It also withdrew an earlier claim about the spectral slope and predicted strong correlation between nearly co-located interferometers [176].

The proposal was contested. Smolyaninov argued that the derivation assumes linear diffraction of Planck radiation, and that vacuum self-focusing would reduce the angular uncertainty to $\sim \lambda_p/L$, “19 orders of magnitude smaller” [177]. Amelino-Camelia called it “a young proposal still looking for

some maturity”. He also stressed that it is unrelated to earlier “holography-inspired” noise estimates of the spacetime-foam type [178], so the phrase “holographic noise” names two different proposals. On the experimental side, GEO600 papers of the period report noise between roughly 100 Hz and 1 kHz that had not yet been explained [179, 180]. The GEO600 team later reported that in “tuned DC readout” experiments “all of the observed noise could be explained” by known sources [180]; that paper does not mention holographic noise. Amelino-Camelia concluded that “the brief season of the ‘GEO600 anomaly’ ... is over” [178]. Hogan’s revised analysis argued that folded arms (GEO600) and Fabry–Perot arms (LIGO) do not amplify holographic noise, so that it was “not currently ruled out by either LIGO or GEO600”. He proposed a dedicated test: cross-correlating two nearly co-located interferometers at frequencies near $c/2L$ [94]. Separately, Kwon and Hogan showed that LIGO/Virgo stochastic-background limits already exclude some *metric*-fluctuation (spacetime-foam) models, whereas their non-metric holographic model remained near the limits of the experiments of the time [181].

Hogan’s most explicitly informational statement is a non-refereed essay. It opens “If reality has finite information content, space has finite fidelity” and posits a “Planck broadcast” bandwidth of about 2×10^{43} bits per second [182]. The essay uses computing metaphors (a “cosmic Internet Service Provider”, the “operating system” the universe “runs on”). It also rejects Planck-size pixels (“nature is not really pixelated in little squares”) in favour of a transverse blur $\langle x_{\perp}^2 \rangle = (2.135 \times 10^{-18} \text{ m})^2 (L/1 \text{ m})$, far larger than the Planck length [182]. It does not propose that the universe is a simulation.

7.3 The Fermilab Holometer

The Holometer consisted of two co-located 39.06 m power-recycled Michelson interferometers (about 2 kW on each beamsplitter, beamsplitters 0.91 m apart, no arm cavities). Their outputs were cross-correlated up to 25 MHz to average down uncorrelated shot noise [4, 183]. Each interferometer had a shot-noise-limited sensitivity of $2.1 \times 10^{-18} \text{ m Hz}^{-1/2}$. After 145 h of data the cross-spectrum reached $2.1 \times 10^{-20} \text{ m Hz}^{-1/2}$, surpassing strain power spectral density t_P for signal bandwidths above 11 kHz [4]. The first analysis tested a “speculative model” of Planckian diffraction from “a putative fundamental Nyquist frequency $f_p = 1/t_p$ ” (the Kwon–Hogan model [181]). It excluded that model at 5.1σ statistical significance (4.6σ including a 10% calibration uncertainty), equivalently a normalization below 44% of the prediction at 95% CL. The authors stressed that this “appl[ies] only to the spectral shape of the particular model” and that the layout “would not respond” to rotational correlations [4]. The final straight-arm analysis (704 h) found no correlation. At 2σ it limited the amplitudes of a two-parameter shear model to $|\beta_L| < 0.10 t_P$ and $|\beta_{2L}| < 0.25 t_P$, well below the order-unity values predicted, and stated explicitly that it does not constrain a newer Lorentz-invariant rotational model [95]. A reconfiguration with one arm of each interferometer bent by 90° tested that rotational model [184] with 1098 h of data. It constrained its normalization to $\eta < 0.25 t_P$ at 95% confidence for one rotation axis [185]. The authors note that the planar geometry cannot test models whose uncertainty entangles all three spatial directions, and that local quantum-field-theoretic models of quantum gravity predict no effect in such a measurement [185]. The same data also gave the first direct limits on MHz stochastic gravitational waves, which lie far above closure density [186].

In short, the Holometer placed null bounds on specific models: two spectral families of shear noise and one rotational model. It did not rule out Planckian information limits in general. A Holometer collaborator later described the first-generation design as resting on a “naive premise” that drew “(justified) criticism” for not being Lorentz invariant, and reinterpreted its null result as “a verification of an exact symmetry” [187]. We could not locate a published version of that

Lorentz-invariance criticism.

7.4 Geotropic fluctuations and next-generation experiments

Verlinde and Zurek started from the covariant entropy bound and postulated “Planck size pixels on the surface bounding a causally connected part of space”. Energy fluctuations of these pixels would produce arm-length fluctuations $\langle \delta L^2 \rangle \sim l_p L$ with strong transverse correlations [188]. Follow-up work tied the effect to the area-law fluctuations of the modular Hamiltonian, computed in AdS/CFT and modelled in flat space by a high-occupation bosonic field [189–191]. Zurek notes that “it is not known how to precisely extend” these ideas to flat space [190]. Li et al. recast published LIGO and Holometer data as rough 3σ bounds on the normalization α (natural value $\alpha \sim 1$). They find $\alpha \lesssim 3$ (LIGO) and $\alpha \lesssim 0.7$ (Holometer) with an infrared cutoff, and $\alpha \lesssim 0.1$ and $\alpha \lesssim 0.6$ without one. Contrary to Hogan’s earlier assumption, they also argue that these fluctuations do accumulate in Fabry–Perot arms [192]. Two new cross-correlation experiments target this regime. GQuEST (Caltech/Fermilab) is a design for 5 m interferometers with photon-counting readout that evades the standard quantum limit. It projects $\alpha < 0.6$ at 3σ in 60 h and $\alpha < 0.1$ in 2160 h at design sensitivity [193]. QUEST (Cardiff) [194] has reported first limits on correlated length fluctuations between 13 and 80 MHz from a 10^4 s run with 1.83 m arms. It frames these as stochastic-background bounds and quotes no constraint on a specific quantum-gravity model [195]. If the effect were real, it would be a large background for Cosmic Explorer and the Einstein Telescope [196]. A tabletop demonstration has shown that quantum-correlated light can outperform classical light in detecting a correlated signal injected into two interferometers in the MHz band [197].

Other theorists dispute that an effect of this size is expected. Carney et al. find that the effective field theory of gravitons “unambiguously predicts” $\Delta L \sim \ell_{\text{pl}} \sim 10^{-35}$ m. On this view a detection would signal “a severe breakdown of effective quantum field theory” [198]. A recent preprint likewise finds graviton-induced noise spectra that are finite but Planck-suppressed [199]. Reviews place these searches among tests of particular phenomenological models of spacetime fluctuations [200, 201].

7.5 Relation to the simulation hypothesis

The logical link is weak and one-directional. (i) A simulation with finite memory must coarse-grain, but nothing requires the coarse-graining to show up as stationary, correlated displacement noise at the MHz frequencies these instruments probe. A simulator could just as well reproduce smooth semiclassical geometry to any accuracy we can measure, so a null result says nothing about simulation as such. (ii) The models that were actually tested (Hogan’s shear and rotational models, the Verlinde–Zurek fluctuations) are derived from holography and quantum-gravity arguments, with normalizations fixed by entropy bounds. None of them is derived from assumptions about a simulating computer. A positive detection would therefore first be evidence about quantum gravity. (iii) None of the theoretical or experimental papers reviewed here invokes the simulation hypothesis [3, 4, 94, 95, 185, 188, 193, 195]. The nearest approach is metaphor: the sampling-theorem language of the Holometer analysis [4] and Hogan’s “Planck broadcast” essay [182]. The only explicit link we found is in a non-refereed preprint. It suggests that if experiments such as the Holometer showed “the holographic universe” to be true, this “may be interpreted as an indication of us participating in a simulation chain” [202]. The experiment it cites [4] reported a null result for one noise spectrum and did not test holography in general. Moreover, holographic scaling, if confirmed, would be at least as naturally read as a fact about quantum gravity as evidence of an “external programmer”. What these experiments do provide is an operational upper bound on some forms of macroscopic, Planck-normalized position noise, and the evidence reviewed here does not

Table 6: Interferometric searches for Planckian, holographically motivated noise. Sensitivities are as quoted by the cited papers (cross-spectrum values unless stated). “Proposal” marks projections, not measurements.

Instrument	Observable	Sensitivity	Model tested	Result	Ref.
GEO600 (folded 600 m arms)	DARM noise, 600 Hz	strain peak ~ 100 – $10^{-22} \text{ Hz}^{-1/2}$ 530 Hz	2.2 $\times$ Hogan holographic noise (comparison, not a dedicated test)	Unexplained noise later attributed to known sources in DC readout; Hogan: not ruled out after arm-folding factor	[3, 94, 180]
LIGO/Virgo	stochastic-background cross-correlation (41.5–169.25 Hz); published strain spectra	$\Omega_{\text{GW}} < 5.6 \times 10^{-6}$ (95% CL) used for foam models	metric-fluctuation (foam) models; geotropic recast	“random walk” model excluded; white-noise model $a_0 < 0.06$; geotropic $\alpha \lesssim 0.1$ (no IR cutoff) or $\lesssim 3$ (IR cutoff)	[181, 192]
Holometer I (39 m, straight arms, 2015–16)	DARM cross-spectrum up to 25 MHz	$2.1 \times 10^{-20} \text{ m Hz}^{-1/2}$ (145 h)	Kwon–Hogan Planckian diffraction; shear-noise family	Excluded at 4.6σ ($< 44\%$ at 95% CL); $ \beta_L < 0.10 t_P$, $ \beta_{2L} < 0.25 t_P$ (2σ , 704 h)	[4, 95]
Holometer II (38.9 m, bent arms)	rotational DARM cross-spectrum, 1.1–20 MHz	$6.1 \times 10^{-21} \text{ m Hz}^{-1/2}$; $\text{CSD}_{\delta h} < t_P/2$ (1098 h)	Hogan–Kwon–Richardson rotational model	$\eta < 0.25 t_P$ (95% CL, one axis)	[185]
QUEST (Cardiff, 1.83 m)	DARM cross-spectrum, 80 MHz	strain 3 $\times$ $10^{-20} \text{ Hz}^{-1/2}$ (10^4 s)	none specific (stochastic length fluctuations)	Upper limits; no model constraint quoted	[195]
GQuEST (Caltech/Fermilab, 5 m)	photon-counting readout, peak $\sim 15.6 \text{ MHz}$	Proposal: nominal signal in several hours	Verlinde–Zurek geotropic fluctuations	Proposal: $\alpha < 0.6$ in 60 h; $\alpha < 0.1$ in 2160 h (3σ)	[193]

support treating them as tests of the simulation hypothesis.

8 Arguments from computational complexity and computability

A second family of arguments asks whether the laws of physics as we know them could be run on a computer at all, and at what cost. Complexity arguments ask how resources scale. Computability arguments ask whether any algorithm can do the job. Before reviewing the literature we separate three tasks that are often conflated:

- (T1) *Execution*: evolving a given initial state forward by the dynamical law, to some accuracy, for a finite time.
- (T2) *Decision*: answering a global question about a model or a family of models, such as whether a ground state has energy below E_0 , whether a spectral gap survives the thermodynamic limit, or whether the system ever halts. Such questions can be intractable for finite instances and undecidable when they quantify over infinitely many times, sizes or instances.
- (T3) *Proof*: deriving every true sentence of a formal theory of the model from its axioms.

A simulator of a universe must perform (T1). Most of the negative results below concern (T2) or (T3). A negative result for (T2) or (T3) bears on the simulation hypothesis only through an additional premise linking it to (T1). We state that premise explicitly in each case.

8.1 Feynman, Lloyd and the extended Church-Turing thesis

Feynman’s keynote address concerned *exact* simulation, in which “the computer will do exactly the same as nature”, and not approximate numerical integration [59]. He imposed the efficiency rule that the number of computer elements should be proportional to the simulated space-time volume. He was candid that this was a stipulation: “I make up the rules, I’m allowed to do that” [59, p. 469]. Under that rule, a classical machine that stores the general probability function of N variables fails, because doubling N requires an “exponentially explosive growth” of the computer [59, p. 472]. His categorical “No!” concerned a different question. He asked whether a *local* classical probabilistic computer can reproduce quantum probabilities. The obstruction he identified is a Bell-type inequality [59, pp. 476, 485]. He conjectured that locally coupled quantum spins could imitate any discrete quantum system. He was confident for bosons but left fermions open [59, p. 476].

Lloyd showed that the conjecture holds for local Hamiltonians. A quantum computer simulates a closed local system with memory and time “directly proportional to the time and space taken up by the system to be simulated” [203]. Continuous variables, such as a lattice gauge field, are handled by discretization, with N qubits per variable giving N bits of accuracy [203]. Bravyi and Kitaev showed that fermionic modes and qubits simulate each other with at most logarithmic overhead per gate [204]. Polynomial-time quantum algorithms exist for scattering in ϕ^4 theory in up to four spacetime dimensions [205]. Lloyd regarded such results as “a strong indication” that any eventual theory of quantum gravity will be efficiently simulable on a quantum computer [52]. That is an expectation, not a theorem.

The complexity-theoretic premise behind “the universe is too hard to simulate” is the failure of the extended Church-Turing thesis (ECT). The ECT holds that any reasonable model of computation can be simulated efficiently by a probabilistic Turing machine. Unlike the original thesis, it is an empirical hypothesis whose truth depends on physics [206]. Bernstein and Vazirani gave the first formal evidence against it: an oracle problem solvable in quantum polynomial time but requiring superpolynomial time classically. They also showed that $\text{BPP} \subseteq \text{BQP} \subseteq \text{P}^{\#\text{P}}$. An unrelativized proof of quantum advantage would therefore require a major breakthrough in complexity theory [207]. Experimental evidence is likewise conditional. Arute et al. estimated that a sampling task their processor performed in about 200 s would take a state-of-the-art classical supercomputer about 10,000 years [208]. A later tensor-network method produced one million samples for that circuit, at fidelity $F \approx 0.0037$, in about 15 hours on 512 GPUs [209].

What follows validly from these results? If the ECT fails in our universe, a *classical* simulator that faithfully reproduces quantum dynamics incurs a superpolynomial overhead on some instances. A quantum simulator does not, by Lloyd’s theorem. The argument therefore constrains the simulator’s architecture and running time. It does not bear on whether a simulation is possible.

8.2 The sign problem

Troyer and Wiese defined a “solution” of the sign problem as a polynomial-complexity algorithm for a thermal average in a model whose related bosonic problem is itself polynomial [210]. They mapped the NP-complete ground-state problem of the three-dimensional $\pm J$ Ising spin glass to a quantum model with a sign problem, $H = -\sum J_{jk}\sigma_j^x\sigma_k^x$. The related bosonic model is a ferromagnet that cluster algorithms simulate efficiently. A *generic* solution of the sign problem would therefore solve an NP-complete problem in randomized polynomial time. The paper states $\text{P} = \text{NP}$, and a footnote correctly refines this to BPP [210]. The result is worst-case. It “does not exclude that a specific sign problem can be solved for a restricted subclass”, and meron-cluster algorithms are an example [210].

Two further points in the paper matter here. First, the hard instances require inverse temperatures growing with N . Classical Monte Carlo on the corresponding spin glass has autocorrelation times $\tau \propto e^{aN}$. Troyer and Wiese also judge that quantum simulators are “most likely not a generic solution” either [210]. Second, physical relaxation towards minimum-energy states, as in spin glasses, soap films and protein folding, is itself subject to local optima and long relaxation times [97]. NP-hardness of the sign problem is therefore a statement about the worst-case cost of a (T2)-type equilibrium task. It does not show that nature efficiently computes something that a computer cannot. A simulator that executes the dynamics (T1) need not solve the spin-glass ground-state problem, just as a physical spin glass does not.

Ringel and Kovrizhin proved a narrower and stronger statement for a specific algorithm class [6]. Take bosonic systems whose gapped phases have a nonvanishing quantized thermal Hall conductance, as most fractional quantum Hall phases do. For these, no *local* sign-free QMC formulation exists. Here “local” means that the configuration variables are obtained from the physical degrees of freedom by finite-depth local quantum circuits. The argument extends to a chiral Kagome antiferromagnet under “additional microscopic assumptions”. The authors state that they “do not address the possibility of nonlocal approaches to QMC, such as determinant QMC or cluster algorithms”. They call obstruction to classical simulation in general “a rather ill-defined task”. They concede that, for non-symmetric transfer matrices, a needed similarity transformation might be nonlocal, “rendering the previous arguments void” [6]. Related results establish intrinsic sign problems for double-semion Hamiltonians [211] and for broad classes of chiral topological phases [212, 213]. Marvian et al. showed that even *finding* a curing transformation within restricted single-qubit classes is NP-complete [214]. None of these results concerns classical algorithms outside the QMC family, or quantum simulators.

The Science Advances article does not mention the simulation hypothesis; we searched the full text for this. Press coverage nevertheless ran headlines such as “Physicists Confirm That We’re Not Living In a Computer Simulation”, asserting that it is “impossible to model the physics of our universe on even the biggest computer” [106]. One report attributed to the paper quotations about memory “built from more atoms than there are in the universe” that do not occur in the published text [215]. Aaronson identified four independent gaps between the paper and the headline. Only local transformations within QMC are excluded. $BPP \neq BQP$ is unproven. A simulator could be quantum. An exponential classical slowdown would be invisible to the simulated [102]. Bostrom made the quantum-computer point in the same context [13].

8.3 Undecidability of physical properties

Cubitt, Perez-Garcia and Wolf constructed translationally invariant, nearest-neighbour Hamiltonians on a 2D lattice of d -level systems, with interaction strength at most 1. Each Hamiltonian is gapped with $\gamma \geq 1$ if a universal Turing machine halts on input n , and gapless otherwise. The spectral-gap problem is therefore undecidable [216, 217]. It follows that for any consistent, recursively axiomatized theory there is such a Hamiltonian whose gap can be neither proved nor disproved [217]. The authors emphasize that “algorithmic undecidability always concerns infinite families of systems”. The property at stake is defined in the thermodynamic limit. Their models can look gapless at all sizes up to a threshold that is itself uncomputable [216]. A review by these authors and co-workers states the general point: undecidability “is a feature of the mathematical model”, and an undecidable problem “must necessarily conceal an infinite somewhere”. At finite sizes it is reflected as complexity-theoretic hardness or as abrupt changes at uncomputably large sizes [218].

Earlier results have the same structure. Moore embedded a Turing machine in the motion of a single particle with three degrees of freedom. For such a system “virtually any question about [its] long-term dynamics is undecidable” even for exactly known initial conditions. Yet finite-time

quantities remain computable, and “the best one can do is to simulate the system and see what happens” [219]. Wolfram put the general point directly. The behaviour of a physical system “may always be calculated by simulating explicitly each step”. Questions about idealized limits of infinite time, volume or precision can be formally undecidable. The finite versions are decidable, though PSPACE-complete for universal cellular automata [220]. Pour-El and Richards exhibited computable initial data for a 3D wave equation whose (weak) solution is not computable [221]. We rely here on the account in Ref. [218]. Weihrauch and Zhong showed that the wave propagator is computable on C^1 data (losing one derivative) and on Sobolev spaces. They argue that the Pour-El-Richards example probably does not yield a wave computer that beats the Turing machine [222]. Wolpert’s limits on *inference devices*, which hold even for super-Turing devices, concern prediction and observation by devices embedded in the universe [223]. On our reading they do not bear on execution by an external machine.

What follows validly is that undecidability of a property (T2) is compatible with, and in the cited constructions is *derived from*, computable local dynamics (T1). These proofs work by embedding a universal Turing machine in a computable model [218]. An argument from undecidability to non-simulability therefore needs a further premise: that executing a single step of the physical dynamics requires the answer to an undecidable question. That premise amounts to hypercomputation in nature.

8.4 The Faizal-Krauss-Shabir-Marino argument

Faizal et al. argue from incompleteness to the impossibility of simulation [7]. Their premises, in order, are as follows. (i) A candidate quantum-gravity theory is a computational formal system $\mathcal{F}_{\text{QG}}$. Its axioms are recursively enumerable; on the same page they are also said to be finite. It is arithmetically expressive and consistent, and satisfies “empirical completeness”. (ii) Gödel’s first theorem gives $\text{Th}(\mathcal{F}_{\text{QG}}) \subsetneq \text{True}(\mathcal{F}_{\text{QG}})$, with truth defined as satisfaction in the standard model of arithmetic. Tarski’s and Chaitin’s theorems add further limits. (iii) “Physically these Gödel sentences correspond to empirically meaningful facts—e.g., specific black-hole microstates.” (iv) Reality instantiates an external, non-recursively-enumerable truth predicate $T(x)$ that is “actualized in nature”. Undecidability results such as the spectral gap supply “empirical backing”, and the Lucas-Penrose argument supplies motivation. (v) Simulation proposals assume “that every physical truth is reducible to the output of a finite algorithm”. They conclude that “any finite algorithm can at best emulate” the computable fragment, that reality “cannot be instantiated on a Turing-equivalent device”, and that the simulation hypothesis is “logically impossible rather than merely implausible” [7]. A university press release described this as showing the idea to be “mathematically impossible” and as possibly “the final, definitive answer” [224].

The valid core is conditional. *If* the observable dynamics of our universe depends on a non-recursively-enumerable predicate, then no Turing-equivalent machine reproduces it. The antecedent is a denial of the physical Church-Turing thesis. Gödel’s theorems do not supply it. Step (ii) concerns provability of arithmetic sentences within a theory, which is task (T3). Step (iii), which carries the argument from (T3) to physics, is asserted without derivation. Redden’s comment makes the corresponding objection. The argument conflates what a formal theory can prove with what an algorithm can execute. A Turing-complete cellular automaton such as Conway’s Game of Life has undecidable long-term questions yet evolves by a finite rule [111]. Non-simulability would require dynamics that must resolve undecidable problems in order to advance, i.e. hypercomputation, for which there is no physical evidence [111]. We add three points. First, the physical undecidability results offered as “empirical backing” are theorems about infinite families of computable models, proved by embedding Turing machines in them (§8.3). They exhibit algorithmic rules generating

undecidable global properties, not non-algorithmic dynamics. Second, even granting all premises, the conclusion excludes only Turing-equivalent simulators. A host universe with hypercomputational resources is not addressed, so “logically impossible” overstates the paper’s own result. Third, premise (v) is not Bostrom’s position: he holds that only what simulated observers could notice need be computed [1]. In a 2026 reply to a published discussion paper, the authors clarified that they take Gödelian truths to be “objectively actualized in physical reality”, independently of human cognition [225]. This restates the contested premise; it does not argue for it. A formal computer-science treatment reaches the opposite verdict under the physical Church-Turing thesis. Using Kleene’s second recursion theorem, Wolpert proved that a universe satisfying the thesis and its converse can simulate itself. He also showed that many questions about simulation are undecidable by Rice’s theorem [89]. The disagreement therefore reduces to whether physical dynamics is Turing-computable. That is an open empirical question, not a consequence of incompleteness.

8.5 Hypercomputation and the physical Church-Turing thesis

The Church-Turing thesis concerns what a human can compute by rote. The physical theses of Deutsch and Wolfram, that every finitely realizable physical system can be simulated by a universal machine, are distinct empirical claims [206]. Idealized Newtonian machines with unbounded speeds violate Gandy’s “Thesis M”. Deutsch argued that a universal Turing machine cannot perfectly simulate continuous classical dynamics [206]. Aaronson argued that Planck-scale physics and the holographic entropy bound appear to exclude Zeno-type and Malament-Hogarth hypercomputers, as well as unlimited-precision analog computation. He suggested that the intractability of NP-complete problems might itself be treated as a physical principle [97]. What matters here is a symmetry. A computability-based argument against simulation requires our physics to violate the physical Church-Turing thesis, and if it does, nothing in the argument prevents the host’s physics from doing the same.

8.6 Resource bounds and the host’s physics

Bostrom estimated $\sim 10^{33}$ – 10^{36} operations for a realistic simulation of human mental history, against $\sim 10^{42}$ operations per second for a planetary-mass computer of known design. He regarded simulating the whole universe at the quantum level as infeasible “unless radically new physics is discovered” [1] (see §2.1). Lloyd bounded the computation the observable universe can have performed at $\lesssim 10^{120}$ operations on $\sim 10^{90}$ bits ($\sim 10^{120}$ with gravitational degrees of freedom). He read these numbers both as upper bounds on computation by matter within the universe and as lower bounds for a quantum computer simulating it directly [52]. Vazza concluded that simulating our universe, or even a low-resolution Earth, is impossible for a simulator in a universe “sharing the same properties”. He added that “only universes with very different physical properties” could produce ours as a simulation, a question he places outside what is scientifically testable [103].

The logical point is that such bounds bind a simulator only if the simulator is subject to our physics. Several authors state this explicitly. Neukart et al. derive that a simulating computer “governed by the same physical laws” would need at least as many particles as the simulated universe, unless the programmer “isn’t limited by the simulation’s physical laws” [202]. Edge and Brown note that Vazza’s conclusions hold “if the simulator’s universe is governed by physics like our own” [226]. Bostrom remarks that “the physics in the basement universe might allow for vastly more powerful computers” [13]. Wolpert’s framework allows simulating and simulated universes to obey different laws [89]. Running time is similarly unconstrained: a slowdown in the host is not observable from inside [102]. Resource arguments therefore constrain two scenarios: Bostrom-type ancestor

Table 7: Complexity and computability arguments: what each establishes about simulability.

Argument	Establishes	Does not establish
Feynman; Lloyd	Exponential cost for classical storage of general quantum states; efficient quantum simulation of local Hamiltonians	That classical simulation is impossible; the ECT’s failure is unproven
Troyer-Wiese	No generic polynomial-time sign-problem solution unless $\text{NP} \subseteq \text{BPP}$	That nature solves these instances efficiently
Ringel-Kovrizhin	No local sign-free QMC for chiral bosonic phases	Anything about nonlocal QMC, other classical algorithms, or quantum simulators
Cubitt et al.; Moore; Wolfram	Undecidable thermodynamic-limit or infinite-time properties of computable models	Uncomputability of finite-time dynamics
Faizal et al.	Incompleteness of r.e. theories (standard)	That physical dynamics is non-algorithmic; exclusion of non-Turing hosts
Bostrom; Lloyd; Vazza	Resource limits for simulators within our physics	Limits on hosts with different physics

simulations run by our descendants, whose physics is ours by hypothesis, and nested simulations within a single physics. They do not constrain the unrestricted hypothesis. The unrestricted hypothesis, in turn, gains immunity from these arguments only at the cost of testability (§4.2).

Table 7 summarizes the section. No complexity-theoretic or computability result reviewed here rules the simulation hypothesis in or out. Taken together, and granting their unproven premises (for example, $\text{BPP} \neq \text{BQP}$), they narrow the admissible designs. A simulator of quantum physics within our physics would have to be quantum or pay superpolynomial overhead. It would have to approximate continuum quantities to finite precision [97, 203]. It would never need to decide the undecidable properties of the models it runs.

9 Rendering on demand, observers, and quantum foundations

A family of proposals holds that a resource-limited simulator would compute (“render”) physical detail only when it is observed, and that quantum measurement is where this shows. We first state what such a test must predict. We then examine the most detailed proposal, by Campbell, Owhadi, Sauvageau and Watkinson [5], and summarise how existing experiments constrain it.

9.1 What a test must predict

The following argument is ours. Suppose a simulator reproduces the outcome statistics of standard quantum mechanics (QM) exactly. Then no experiment whose statistics QM predicts can distinguish it from a world that simply obeys QM. This includes the delayed-choice and quantum-eraser experiments often invoked as support. Their surprising features, such as a pattern that depends on a later choice of measurement, are QM predictions, not anomalies. A render-on-demand hypothesis becomes testable only when it names an observable (a fringe visibility V , a conditional probability) and predicts a deviation Δ from the QM value that exceeds the experimental uncertainty. Experiments that agree with QM then bound Δ (Section 9.3).

9.2 The proposal of Campbell et al. (2017)

Assumptions and conjecture. Campbell et al. assume that the simulating system “is finite” and “seeks to achieve low computational complexity”. By analogy with procedurally generated games, they infer that it “would render reality only at the moment the corresponding information becomes available for observation by a conscious observer” [5, p. 80]. To accommodate Bell violations, they posit that locality inside the simulation does “not constrain the action space of the system performing the simulation” [5, p. 81]. The simulator is therefore explicitly nonlocal with respect to simulated spacetime. The testable conjecture is that wave or particle patterns are fixed “by the existence and availability of the which-way data when the pattern is observed”, not at detection [5, p. 82]. “Availability” means recording “on objective media” [5, p. 87]. The paper does not say when a physical record, such as an atomic state or an emitted photon, counts as one.

Proposed experiments. The paper proposes four schemes:

1. which-path detectors are left on at the slits, but the recorder is unplugged; success is interference. In the entangled version, the coincidence counter of the Kim et al. eraser [227] is removed and “D0 should display the wave pattern if the experiment is successful” [5, p. 86];
2. which-path and screen data go to separate USB drives, and the which-path drive is irreversibly destroyed with probability $1/2$; success is interference only where the partner drive was destroyed [5, pp. 87–88];
3. in a delayed-choice eraser, the signal position X predicts the later random which-path/erasure outcome R via $P[R=1|X \approx x] \approx [1 + 2 \cos^2(\pi x/a)]^{-1}$ with $a = \lambda L/d$ [5, Eq. (2)];
4. a thought experiment with a manual switch and delays of about 60 s. A total-variation argument excludes X being drawn from the wave distribution with the switch off and from the particle distribution with it on [5, Eqs. (3)–(4)].

The authors “cannot predict the outcome” of scheme 4 but prove it “will be new” [5, p. 83]. No scheme carries a numerical prediction. The only tolerance given is a noise allowance $\delta(I) < 0.9$ in scheme 4 [5, p. 94].

Do the schemes predict deviations from QM? They do, although the paper does not compare its success criteria with QM. The comparison below is ours and draws on the cited literature. In QM, which-path information carried by any system suppresses interference, to the degree that the carrier’s states are distinguishable, whether or not anyone reads it. The information “is still present in the universe, independent of whether an observer takes note of it or not”, and ignoring its carrier leaves a mixed state “which cannot show an interference pattern” [228, Sec. II.E]. Schemes 1 and 2 therefore demand visibilities of order unity where QM bounds V by the which-path parameter I , through $I^2 + V^2 \leq 1$ [229, 230]. In QM, destroying a classical copy of the record does not restore interference. Interference reappears only conditionally, through a suitable measurement on the quantum carrier of the path information [228, 231].

In the Kim et al. eraser, D0 coincidences with the two erasing detectors give fringes and antifringes ($R_{01} \propto \cos^2$, $R_{02} \propto \sin^2$). The which-path coincidences give none [227, Eq. (10)]. The unsorted D0 data are thus fringe-free, “just noise” in Kastner’s words [231]. Removing the coincidence counter (scheme 1) therefore cannot produce fringes in QM. The erasure subensemble (pooling the two erasing detectors) and the which-path subensemble are both fringe-free, so in QM X and R are independent and $P(R=1|X) = 1/2$. Eq. (2) instead assumes a single-detector fringe for $R = 0$.

Table 8: Experiments that bound render-on-demand deviations (V : fringe visibility; I : which-path parameter; uncertainties as quoted, $\pm 1\sigma$ where stated).

Experiment	Varied, or left unread	Result
Jacques et al. [237]	Open/closed choice by QRNG, space-like separated from entry (48 m)	Closed: $V = 94\%$; open: no fringes, port probabilities 0.50 ± 0.01 , $I > 0.99$
Kim et al. [227]	Idler registered ≥ 8 ns after the signal	Fringes/antifringes with D1/D2; none with D3/D4
Ma et al. 2013 [230]	Erasure choice space-like separated (55 m, 144 km), or $\approx 450 \mu\text{s}$ later	$I = 0.955(7)$, no fringes; erasure: $V = 0.951(18)$, $I = 0.077(22)$
Ma et al. 2012 [238]	Swapping choice 14–313 ns after the other photons were registered	Fidelity 0.681 ± 0.034 vs 0.421 ± 0.029
Dürr et al. [239] (via [228])	Path stored in atomic internal states, not read out	Fringes vanish; momentum kick 10^{-4} of fringe period
Hackermüller et al. [240]	Path carried off by undetected thermal photons	$V = 47, 29, 7, 0\%$ at 0, 3, 6, 10.5 W; matches decoherence theory
Hornberger et al. [241]	Path left in colliding gas molecules	Pressure-dependent loss of V , as predicted

It departs from QM by up to $1/2$ in probability at dark fringes, where it gives ≈ 1 , and by $1/6$ at bright fringes, where it gives $1/3$. Scheme 4 assumes “an interference pattern at D0” in its configurations (a)–(c) [5, p. 91]. That is also not a QM prediction for an entangled-pair eraser, where fringes appear only after conditioning on the partner photon [230, 231]. The paper classifies the possible outcomes as a “discontinuity in the rendering reality”, rendering at observation, or a paradox [5, p. 91]. Under this scheme, the QM result (no fringes in unconditioned D0 data in any configuration) counts as a “discontinuity”. In configuration (d) it coincides with the outcome the authors call “a strong indicator that this reality is simulated” [5, p. 92]. As written, the protocol has no outcome that counts against the hypothesis.

Status. The non-profit CUSAC, founded by Campbell, lists five experiments “that CUSAC intends to perform”. It names leads in California (F. Khoshnoud) and Canada (D. Chartrand) but posts no data [232]. A paper co-authored by Khoshnoud, which acknowledges Campbell and Chartrand, reports a teaching-lab single-photon double slit with $|V| = 0.65 \pm 0.22$. It also reports a polarisation eraser in which orthogonal polarisers on the slits removed the fringes without any measurement: “we did not need to measure the path information” [233]. That paper is not presented as a test of the simulation hypothesis, and a polarisation marker is not the unplugged recorder of scheme 1. As of September 2026 we found no citable report of any of the four schemes having been performed.

9.3 What existing experiments constrain

Wheeler proposed choosing the measurement after the photon has entered the interferometer [234], and later described a cosmological version [235]. His criterion for a completed phenomenon was an “irreversible act of amplification”, not conscious awareness [228, Sec. II.D]. The quantum eraser [236] extended the scheme to entangled pairs. Table 8 lists realisations that test the two questions a rendering hypothesis raises: does the timing of the choice matter, and does an unread record matter? All agree with QM within their quoted uncertainties. The review by Ma, Kofler and Zeilinger concludes that “no physical interactions or signals, let alone into the past, are necessary to explain the experimental results” [228, Sec. VI].

Two limits apply. First, the delayed-choice experiments vary the timing of the choice. None of them destroyed a classical record, or varied whether a person viewed it, as schemes 1–2 require. Second, the decoherence and internal-state experiments show interference lost to unread records. Dürr et al. put this as “the mere fact that which-path information is stored in the detector and could be read out already destroys the interference pattern” [228, Sec. IV.B]. Because “availability” is not operationally defined, however, a proponent could call such records “available”. On our reading, any deviation tied to *when* a choice is made is bounded at the few-per-cent level of the best photonic V and I uncertainties. A test of the unread-record variant requires fixing the availability criterion before the data are taken.

9.4 Other observer-dependent and “glitch” proposals

Whitworth argued, in a technical report, that a virtual world “would likewise be expected to be calculated only on demand” [242]. Arvan proposed a “peer-to-peer simulation hypothesis” for quantum phenomena [243]. Irwin, Amaral and Chester’s “self-simulation” interpretation [244] asserts that it “became more widely agreed that it is consciousness”, rather than interaction with a detector, that changes the double-slit pattern. This is at odds with the review literature above [228]. As evidence they cite Radin et al.’s attention experiments [245]. An independent reanalysis of the 2016 dataset found shifts in the predicted direction but “not deemed significant ($p > 0.05$)”, and concluded that it “does not contain evidence of mind-matter interaction” [246]. Yampolskiy’s “How to hack the simulation?”, published as “How to Escape From the Simulation” [247], does not evaluate evidence. It proposes that “every novel QM experiment can be seen as an attempt at hacking the simulation”, and it surveys proposals to overload the simulator with computationally intense activity. A reviewer comment printed with the paper argues that if simulators cannot optimise through layers of abstraction, a full-universe simulation is more likely. That would make it futile to expand “the human-observed-in-detail cone” [247]. None of these works gives a quantitative prediction that differs from QM. By contrast, Chalmers and McQueen give explicit dynamics for consciousness-triggered collapse and report that simple versions are “falsified by the quantum Zeno effect”. More complex versions remain compatible with current data and are in principle testable with quantum computers [248]. Their model is not a simulation proposal, but it shows the kind of formulation the render-on-demand literature lacks.

9.5 Nonlocality, superdeterminism and freedom of choice

Loophole-free Bell tests constrain any simulator that works from local data. Hensen et al. found $S = 2.42 \pm 0.20$ in 245 trials at 1.3 km ($p = 0.039$) [249]. Giustina et al. reported $p \leq 3.74 \times 10^{-31}$ [250]. Shalm et al. reported p down to 5.9×10^{-9} , or 2.3×10^{-7} after adjusting for setting predictability [251]. A simulator matching these results must therefore act nonlocally with respect to simulated spacetime, as Campbell et al. posit, or correlate its hidden state with the settings. The latter is a violation of Statistical Independence, i.e. superdeterminism [252]. Campbell et al.’s “VR engine reacting to the intention of the experimentalist” [5, p. 94] can be read as falling in this class.

In-universe versions of the setting-correlation option have been narrowed. The BIG Bell Test used 97,347,490 choices from about 100,000 participants [253]. Cosmic Bell tests used Milky Way stars, with violations of 7.31σ and 11.93σ quoted as lower bounds, pushing the latest local-realist intervention back by ~ 600 yr [254]. With quasars they reached 9.3σ and pushed it back to at least ~ 7.8 Gyr ago, covering 96% of the past-light-cone space-time volume [255]. The cosmic tests assume fair sampling, and all of these tests constrain *local* mechanisms. Hossenfelder and Palmer argue that they “do not – cannot – rule out Superdeterminism”, and that “one cannot

prove freedom of choice by assuming freedom of choice” [252, Sec. 4.3]. Neither kind of simulator above is constrained by these bounds. A nonlocal one is not, because Bell tests exclude only local models. One that sets settings and outcomes jointly from outside spacetime is not either, because it is empirically a superdeterministic model. It therefore inherits superdeterminism’s testability problem: the signature Hossenfelder and Palmer propose, Born-rule-violating time-correlations for nearly identical preparations, needs a concrete model to fix how similar the preparations must be [252, Sec. 6]. ’t Hooft uses a computer image himself: a God who must “run a universe” prefers deterministic rules on a classical computer. He sees “no objections against super-determinism” [256]. His cellular automaton interpretation predicts that quantum computers cannot outperform a classical computer with one memory site per Planck volume updating once per Planck time. He states that exceeding this bound would falsify the theory [74, Sec. 5.8]. That is a quantified, resource-based prediction of the kind the render-on-demand proposals reviewed here do not yet provide.

10 Information-theoretic proposals: the mass of information and “infodynamics”

One reading of the simulation hypothesis holds that if the world is computed, information should behave as a physical quantity with dynamics of its own. The most widely publicized proposals in this vein are by M. M. Vopson: a *mass–energy–information* (M/E/I) equivalence principle and a *second law of information dynamics* (“infodynamics”). The second law is presented as evidence that “appears to underpin” the simulated-universe hypothesis [8]. This section keeps three things apart: (i) the established thermodynamics of information, (ii) the conjectures built on it, and (iii) what would test those conjectures, and what the tests show so far. Statements labelled “our calculation” are our own checks. The accompanying research notes give the details needed to reproduce them.

10.1 Established background: Landauer’s principle

Landauer argued that logically irreversible operations are associated with physical irreversibility and require a minimal heat generation “typically of the order of kT for each irreversible function” [257]. His central calculation concerns resetting a thermal ensemble of bits. This lowers their entropy by $k \ln 2$ per bit, so $0.6931 kT$ per restored bit must appear as heat “to the surroundings” [257]. Three further points in the same paper matter below.

- Devices such as a particle in a symmetric bistable well “can hold information without dissipating energy”.
- Resetting bits that are already known to be in the target state involves “no entropy changes” and “no heat dissipation”.
- When stored information thermalizes, the entropy of the information-bearing degrees of freedom “can increase by $kN \log_e 2$ ”.

Bennett’s formulation makes the bookkeeping explicit [258]. An entropy decrease of the information-bearing degrees of freedom must be compensated by an entropy increase elsewhere, typically as “energy imported into the computer, converted to heat, and dissipated into the environment”. Erasing random data can be thermodynamically reversible, whereas erasing known data is thermodynamically irreversible [258].

The bound has been tested in several systems:

- *Colloidal bead, double-well optical trap.* The mean dissipated heat saturates at the Landauer bound for long erasure cycles [259].
- *Colloidal particle, feedback trap.* The extrapolated mean work for full erasure is $0.71 \pm 0.03 kT$, compared with $\ln 2 \approx 0.693$, and $0.05 kT$ in a control protocol that does not erase. The authors note that this work “is dissipated into a surrounding heat bath” [260].
- *Nanomagnetic bits.* The dissipation is $(4.2 \pm 0.9) \text{ zJ}$, i.e. $(1.0 \pm 0.22) k_B T$ at 300 K, within two standard deviations of $k_B T \ln 2$ [261].

Modern treatments place these results within stochastic thermodynamics [262]. In that framework the nonequilibrium free energy is $F = E - TS$, with S the Shannon (von Neumann) entropy. It exceeds its equilibrium value by $T D[\rho||\rho_{\text{eq}}]$, the temperature times the relative entropy of the actual state with respect to equilibrium [263]. The principle’s derivations have critics: Norton argues that they rest on incorrect assumptions and neglect fluctuations [264]. None of the proposals below turns on that dispute, however. We take the Landauer bound, with the heat going to the environment, as the established baseline.

10.2 The mass–energy–information conjecture

Claims. Vopson proposed that a stored bit has a rest mass

$$m_{\text{bit}} = k_B T \ln 2 / c^2, \quad (5)$$

which gives $3.19 \times 10^{-38} \text{ kg}$ at 300 K. He proposed a test: a data storage device should be heavier when full than when erased, by $2.5 \times 10^{-25} \text{ kg}$ for 1 Tb [265]. The rationale is a syllogism: “if $M = E$ and $E = I$, then $M = E = I$ ”. On this view the Landauer energy of a bit “condenses into its equivalent mass-energy when the information is stored at equilibrium” [10]. The same numerical value appeared earlier with a different meaning. Herrera attached $\approx 3 \times 10^{-35} \text{ g}$ to one bit as the *decrease* of a system’s mass that accompanies the energy dissipated when a bit changes [266].

Later papers extend the idea in two steps.

- *Information per particle.* Vopson takes the Shannon entropy of the relative abundances of particle species in baryonic matter as the “information content per particle”. This gives 1.288 bits for $\{p, e, n\}$ and 1.509 bits for $\{u, d, e\}$, and about 6×10^{80} bits in the observable universe [267]. He postulates that stable massive particles store this information “about themselves” [267].
- *Wider consequences.* An earlier estimate tied $\sim 5 \times 10^{94}$ bits to the missing dark matter; the 2021 paper itself calls this “most likely overestimated” [267]. Missing information has also been reviewed as a dark-matter candidate [268]. Growth of digital data was projected to exceed half of Earth’s mass in ~ 500 years [269].

The proposed test. The experimental protocol combines these ingredients [270].

- *Temperature-dependent mass.* A body at temperature T should carry an information mass. For 1 kg of copper changed by 100 K the prediction is $\Delta m_{\text{inf}} = 3.33 \times 10^{-11} \text{ kg}$. The paper states that the “physical mass” does not change with temperature.
- *Annihilation photons.* Because the electron’s putative information mass is $\sim 2 \times 10^7$ times smaller than its rest mass, the paper instead proposes to “erase” it by e^+e^- annihilation. In

addition to the two 511 keV photons, two photons of wavelength $\lambda = hc/(Ik_B T \ln 2)$ should appear. This is $\sim 50 \mu\text{m}$ at room temperature for $I = 1.509$ bits and $25\text{--}75 \mu\text{m}$ for 1–3 bits. We compute $45.9 \mu\text{m}$ for $I = 1.509$ at 300 K, i.e. 0.027 eV per photon.

- *Apparatus.* A ^{22}Na source, a W(100) moderator, a nm-thin Al target and temperature control. The predicted linear shift of λ with T is the main control.

The protocol states its assumptions openly. It assumes that the annihilating particles share the target temperature, and it makes “a strong assumption” that the information energy leaves as IR photons rather than, for example, with the gamma rays. Hence it is “not totally guaranteed to succeed” even if the conjectures are correct [270]. A null result would therefore not falsify M/E/I.

Status of tests. The proponent states that M/E/I “still awaits an experimental confirmation” [10] and “has no empirical validation yet” [271]. We are not aware of any reported execution of the annihilation protocol. The only direct weighings of storage media that we found had an accuracy of 10^{-8} kg. They recorded unexplained transient weight changes of order 1 mg, which their authors did not attribute to a mass of information [272, 273]. That accuracy is about 17 orders of magnitude short of the predicted 2.5×10^{-25} kg per terabyte.

Critiques and assessment. Two published critiques address the conjecture directly. Burgin and Mikkilineni argue that only physical *representations* of information have mass, and that these can differ for the same information [274]. Lairez argues on thermodynamic grounds [275]. At constant temperature the internal energy of a system of independent entities is constant, so an entropy change $T\Delta S$ is exchanged with the surroundings and cannot be localized as rest mass. He adds that erasing the same bit twice, or erasing after making an independent copy, would expose the inconsistency of attributing mass to logical information. A 2024 review of the Landauer bound records these objections [276]. Lairez also disputes Landauer’s principle itself, but the mass argument does not depend on that dispute.

Our assessment agrees with Lairez’s thermodynamic point and rests only on the standard results above.

1. *A stored bit in a symmetric memory carries no extra rest mass.* Landauer’s symmetric memory stores a bit with no energy difference between its logical states. Kish and Granqvist make the same point for such memories: the system’s “energy and mass ... remain the same” after writing [272]. A bit known to be in one well therefore has the same mean energy E as the thermalized bit. It differs only in nonequilibrium free energy, by $TD[\rho||\rho_{\text{eq}}] = k_B T \ln 2$ (Ref. [263]; the value $\ln 2$ is our evaluation for a symmetric bit). Since the mass of a body measures its energy content, $m = E_0/c^2$ [277], this entropic free-energy excess implies no extra rest mass. The $k_B T \ln 2$ that appears on erasure is supplied as work by the eraser and leaves as heat [258, 260]. It is not released from the bit’s rest energy. For asymmetric memories a mass difference of either sign does exist, but it is set by the device’s energy asymmetry [272], not by $k_B T \ln 2$.
2. *The copper prediction already conflicts with calorimetry (our calculation).* Take the conjecture literally, including $E = mc^2$ and energy conservation. Then the temperature-dependent Δm_{inf} of the protocol [270] must be supplied as energy when the body is heated. For copper this is an extra heat capacity of $1.509 \times 219.5 k_B \ln 2$ per atom, i.e. $1.9 \text{ kJ mol}^{-1} \text{ K}^{-1}$. That is about 78 times the measured $C_p \approx 24.5 \text{ J mol}^{-1} \text{ K}^{-1}$, which we obtain from the NIST-JANAF parameters [278]. Equivalently, $\Delta m_{\text{inf}} c^2 \approx 3.0 \text{ MJ}$ for the 1 kg, 100 K example, whereas the

measured enthalpy increment between 300 and 400 K is 3.9×10^4 J. The ordinary relativistic mass increase of the heated block, $\Delta H/c^2 \approx 4 \times 10^{-13}$ kg [277], is non-zero and about 80 times smaller than the claimed information mass.

3. *The “information per particle” is a property of the mixture (our calculation).* The 1.288 and 1.509 bits are entropies of the species mix in a population, not properties of a single particle. A pure-hydrogen population would give exactly 1 bit. The predicted IR wavelength inherits this dependence.

10.3 The “second law of infodynamics”

Claims. Vopson and Lepadatu define the entropy of “information-bearing states” as

$$S_{\text{inf}} = Nk_B \ln 2 H(X), \quad (6)$$

where N is the number of information states and $H(X)$ is the Shannon entropy of the symbol frequencies in the stored string. For example, $H = 0.991$ for the 88-bit ASCII encoding of “INFORMATION” [10]. They assert $\partial S_{\text{inf}}/\partial t \leq 0$ and give two examples.

- *Digital storage.* In a micromagnetic Monte Carlo simulation, written bits self-erase by thermal relaxation. N , and with it S_{inf} , falls. The decrease is said to be compensated by a rise in physical entropy.
- *SARS-CoV-2 genomes.* Ten genomes of identical length (29,903 nt) were selected from NCBI because they “displayed an incremental number of SNP mutations with time” (0–49 over 22 months). Their nucleotide-composition entropy falls from 1.9570243 to 1.9562614 bits, “rather linearly” in time [10].

The authors read the genomic trend as hinting at a deterministic, entropy-driven mutation process [10, 279].

Later work extends the law to atomic physics (Hund’s rules), to the prevalence of symmetry, and to cosmology [8, 280]. The cosmological derivation assumed $dQ = T dS = 0$ for adiabatic expansion. After criticism that this holds only for reversible processes, Vopson proposed a modified argument with added “loss” terms [281]. Finally, a toy model of Planck-area cells storing 0 (empty) or 1 (occupied) is used to argue that gravitational clustering lowers H . It derives $F = GMm/R^2$ by combining $\Delta S_{\text{inf}} = k_B \ln 2 H$ per step $\Delta r = \hbar/mc$ with $M = NHk_B T \ln 2/c^2$ and $N \approx R^2/L_P^2$. On this basis gravity is presented as “data compression” that “supports the possibility of a simulated or computational universe” [280].

Independent analyses and critiques.

- *Replication with large datasets.* Ostrowski *et al.* fitted composition-entropy trends to 8.8 million SARS-CoV-2, 1.1 million HIV and 1 million influenza records and found negative slopes throughout [282]. For SARS-CoV-2 the slope is $(-0.972 \pm 0.018) \times 10^{-5}$ per mutation by least squares, with $\chi^2 = 7.6 \times 10^7$, and $(-1.146 \pm 0.421) \times 10^{-5}$ by robust Bayesian regression. The robust slopes for HIV and influenza lie within 1.4σ of zero (our reading of their Table I). The authors interpret the trend through nonequilibrium thermodynamics rather than a new law, and do not test a mutational-bias null model.

- *Reformulations.* Crecraft attributes the magnetic example to the ordinary second law. He argues that the statistical information entropy used is “ill defined” as the basis of a physical law, and notes that unbiased random mutations “would be expected to increase” it. He then proposes a substantially different “thermocontextual” law [283]. Todd reinterprets the decreasing quantity as a relative entropy to equilibrium, whose decrease is compatible with the second law [284].
- *Other applications.* Kubiński *et al.* report a U-shaped (eventually increasing) Shannon entropy in simulated nuclear fuel [285].
- *Public critique of the gravity paper.* Hossenfelder argued in a video that H is unchanged by swapping 0s and 1s and is maximal at equal numbers, and that the paper “shouldn’t have been published” [286], as transcribed in Ref. [271]. Vopson replied that the video gives no explicit mathematical rebuttal and that fixing an encoding gives bit values meaning [271]. The reply and the cosmological note appeared as “News and Views” items in *IPI Letters*, a journal edited by Vopson. The journal states that such items are screened at editorial level rather than going through the usual peer review.

Assessment (our analysis).

1. *Genome data.* We downloaded the ten accessions and reproduced every published entropy to seven decimals, using the plug-in estimator on A/C/G/U counts. The decrease is then a direct consequence of the known mutational spectrum.
 - Moving one site from base a to base b changes H by $\approx N^{-1} \log_2(p_a/p_b)$, which is -2.7×10^{-5} bits for C→U.
 - SARS-CoV-2 is U-rich and C-poor (32.1% vs 18.4%), and 38–42% of its early changes are C→U, attributed to host APOBEC-like editing [287].
 - In the most mutated genome of the table, 22 of 49 differences are C→U, and 41 of 49 move a site toward a more common base.
 - A null model with unbiased substitutions predicts an *increase* of $+3.9 \times 10^{-6}$ bits per substitution, compared with the observed mean of -1.6×10^{-5} bits per SNP.

The choice of estimator is not the issue here. At fixed N the Miller–Madow correction is a constant offset [288]. What matters is the absence of a null model and the selection of sequences “with an incremental number” of SNPs. Simmonds attributes the U-rich, C-poor composition of other human coronaviruses to prolonged C→U hypermutation [287]. A composition drift of this kind is what a biased mutation process produces, and it does not indicate a new law.

2. *Digital example.* For the magnetic medium, Landauer’s accounting says that thermalization *raises* the entropy of the information-bearing degrees of freedom by up to $Nk_B \ln 2$ [257]. What falls is the relative entropy the medium retains about the written record. For undriven relaxation, $\Delta I = -\Delta_i S \leq 0$ follows from Eq. (12) of Ref. [263]. The “second law of infodynamics” in this example is therefore a restatement of the ordinary second law, as Crecraft and Todd also conclude [283, 284].
3. *Gravity model.* In the gravity model H depends only on the counts (N_0, N_1) . It is unchanged when a particle moves through empty cells and changes only when particles merge. The $1/R^2$ law enters through the identifications $\Delta r = \hbar/mc$, $M = N\hbar k_B T \ln 2/c^2$ and $N \approx R^2/L_P^2$, not through a configuration-dependent entropy [280]. Moreover, it uses the M/E/I relation, whose difficulties are discussed above.

Table 9: Status of the information-theoretic claims discussed in Sec. 10.

Claim	Status	Decisive test	Test status
Landauer bound, $\geq k_B T \ln 2$ of heat per erased bit [257]	Established; derivations debated [264]	Single-bit erasure experiments	Confirmed within errors [259–261]
Stored bit has rest mass $k_B T \ln 2/c^2$ [265]	Conjecture; contested [274, 275]	Weighing a full vs. erased memory ($\sim 10^{-25}$ kg/Tb)	Not performed; best weighings are $\sim 10^{-8}$ kg [272]
Temperature-dependent information mass of matter [270]	Conjecture	Mass or heat capacity vs. T	In conflict with calorimetry (our calculation)
$\sim 50 \mu\text{m}$ photons from e^+e^- annihilation [270]	Prediction, with a stated escape clause	Coincident far-IR detection with T -dependent λ	No experiment reported
Second law of infodynamics [10]	Proposed law; reinterpreted [283, 284]	Null models for mutation bias; entropy accounting	Trends reproduced [282]; explained by mutational bias and the ordinary second law (our analysis)
Gravity as information compression [280]	Conjecture; criticized informally [286]	None proposed	—

10.4 Summary

Table 9 summarizes the status of each claim. The empirically established core, Landauer’s bound with heat delivered to the environment, gives no support to a rest mass of stored information. As formulated, the M/E/I conjecture conflicts with the standard energy bookkeeping of symmetric memories. Its temperature-dependent version conflicts with calorimetry by nearly two orders of magnitude (our calculation). Its flagship test has not been performed, and it has a declared escape route if no photons are seen. The infodynamics trends are reproducible arithmetic on well-defined data. Their sign is explained by a known mutational bias in the genomic case and by the ordinary second law in the digital case. On the present evidence, therefore, neither proposal provides empirical support specific to the hypothesis that the universe is computed, contrary to the framing in Refs. [8, 280]. The infodynamics analyses are cheap to audit, and the research notes outline a protocol for doing so.

11 Discussion and open problems

11.1 What is constrained

Table 2 summarizes the status of each restricted sub-hypothesis. Three patterns emerge from Secs. 5–10.

First, the only quantitative empirical constraints concern *preferred-frame discreteness* and one specific *holographic-noise* model. For a hypercubic lattice with unimproved Wilson fermions, existing Lorentz-violation limits bound the inverse lattice spacing far above the cosmic-ray estimate of the original proposal (Sec. 5.4). The Holometer excluded the tested holographic-noise model [4]. Both constraints fall on the physics of discreteness, whether or not it is simulated. They are evaded by Lorentz-invariant discreteness [93], by improved discretizations (Sec. 5.4), and by noise models with a different spectral or spatial structure (Sec. 7).

Second, the proposals aimed at the *simulation* rather than at discreteness are either untested or,

as stated, inconsistent with established physics. The render-on-demand schemes count as “success” an outcome that standard quantum mechanics excludes, and none has been carried out (Sec. 9). The information-physics proposals have one sharp, unperformed prediction. The empirical trends reported for them have standard explanations (Sec. 10).

Third, the complexity and computability arguments on both sides depend on premises that do the real work. Examples are that the simulator obeys our physics, that it is a Turing-equivalent machine, or that it must reproduce every property of a model rather than evolve it for a finite time (Sec. 8). None of these arguments is an empirical test.

11.2 Open problems

We list open problems that we consider tractable, in order of feasibility.

1. *A cubic-harmonic anisotropy search.* The O_h -invariant $\ell = 4$ and $\ell = 6$ analysis specified in Sec. 6.7 can be carried out with public data. Section 5.4 shows that such a search is informative only for simulators whose $O(b^2)$ artefacts are removed to all orders and whose b^{-1} lies near 10^{11} GeV. A null result would therefore bound a narrow but well-defined design class. A detection would be evidence for discreteness, not for simulation. A prerequisite is a theory prediction of the anisotropy amplitude, as a function of b , after propagation and magnetic deflection. No such prediction exists yet (Sec. 6).
2. *Lattice-theory checks of the dispersion mapping.* Our derivations in Sec. 5.4 use isotropic effective-field-theory bounds and free-field dispersion relations. Three extensions would make them rigorous: direction-dependent Standard-Model-Extension limits, the photon-decay rate in the lattice theory itself, and dispersion relations for domain-wall and overlap fermions.
3. *A quantified render-on-demand model.* We found no model that predicts a *quantified* deviation from quantum mechanics as a function of a simulator’s resource budget (Sec. 9). Without one, no experiment can test the idea. With one, existing delayed-choice and decoherence experiments would already bound the budget.
4. *Tests of the mass–energy–information conjecture.* The proposed annihilation experiment [270] and the calorimetric consistency argument of Sec. 10 give definite targets. We found no reported attempt at the former.
5. *Formal treatment of nesting and host physics.* The Bayesian treatment of the simulation argument [27] could be extended to nested simulations with resources that decrease with depth [26], and to simulators whose physics differs from ours. This would clarify which conclusions depend on assuming that the simulator shares our physics.

11.3 Limitations of this review

Our ledgers record every source that we could read only as an abstract or a bibliographic record. The main gaps are the following:

- Several philosophy papers were paywalled.
- The primary sources of Zuse, Fredkin, Greisen and Zatsepin–Kuzmin could not be consulted. Claims about them are attributed to the secondary sources that report them.

- Some bibliographic databases could not be queried automatically (PhilPapers, NASA ADS full text, Google Scholar). Every “we are not aware of” statement in this paper refers to the documented searches in the ledgers, which did not include these databases. We will repeat these searches manually before publication.

The analyses labelled “this work” have been checked numerically. They have not been checked by independent experts, and we invite scrutiny of them in particular.

Data and code availability. The claim ledgers, per-section research notes, derivations, the numerical check of the lattice photon-decay threshold, the SARS-CoV-2 entropy reproduction scripts, and the citation-checking tool are available in the public repository accompanying this paper.

11.4 Conclusion

As a claim about reality, the simulation hypothesis cannot be settled by observation. Specific hypotheses about how a simulator would be built can be tested, and some already have been constrained, although often more strongly than their proponents noted. The most informative future work will not ask whether the universe is simulated. It will ask which classes of discrete, resource-bounded or information-physical dynamics nature permits. That question belongs to physics whatever the answer to the first.

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